Università di Catania

Presentation Attacks in the Era of *AI Forensics*

Stefano Marrone
University of Naples Federico II

Università degli Studi di Napoli Federico II, Italy

Università degli Studi di Cagliari, Italy

University of Catania, 18 March 2026

What is Artifi

- Marvin Minsky de   nes capable of doing autonomously thi   ngs"
- In 1700, Wolfgan   ne capable of playing chess

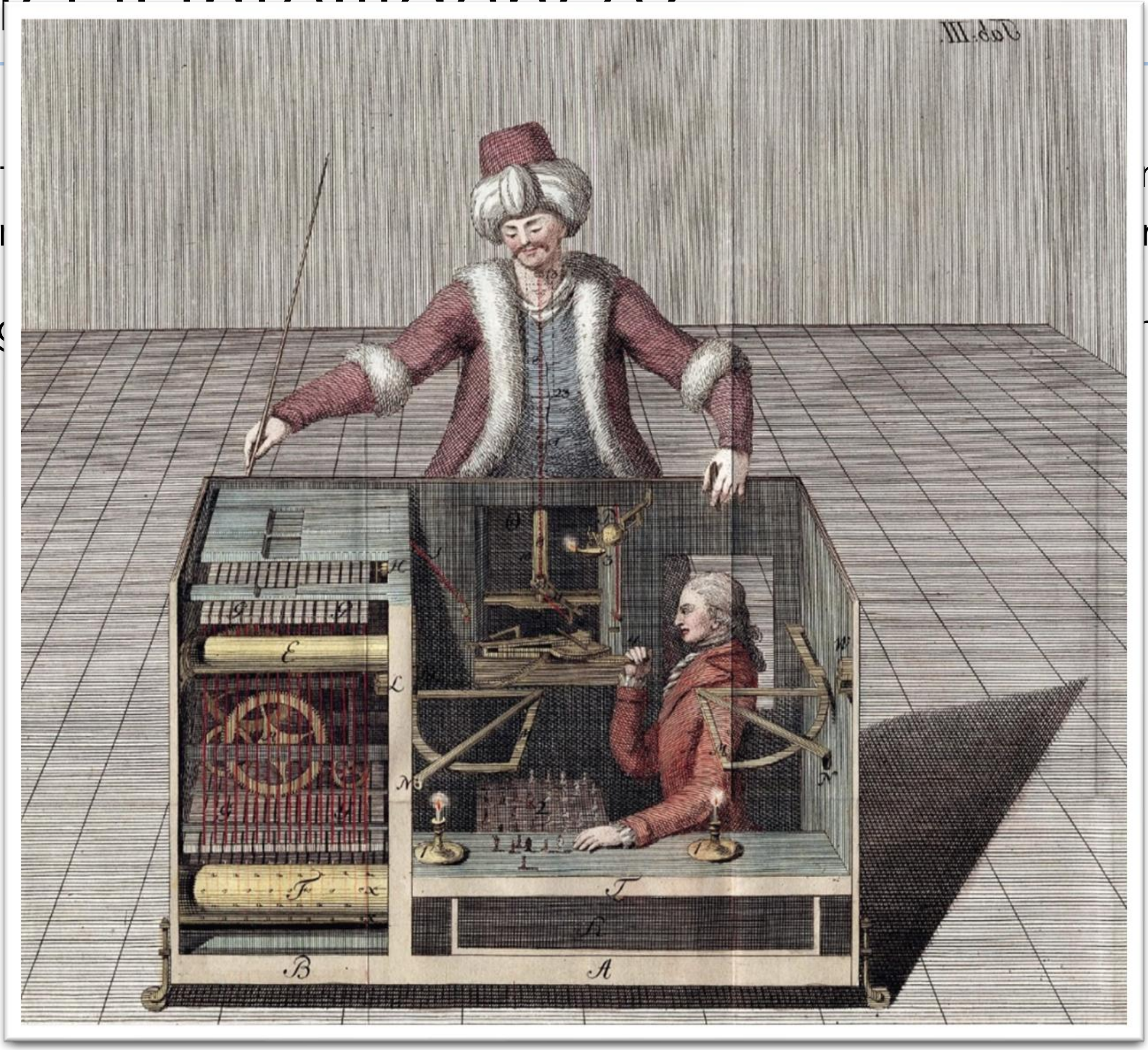
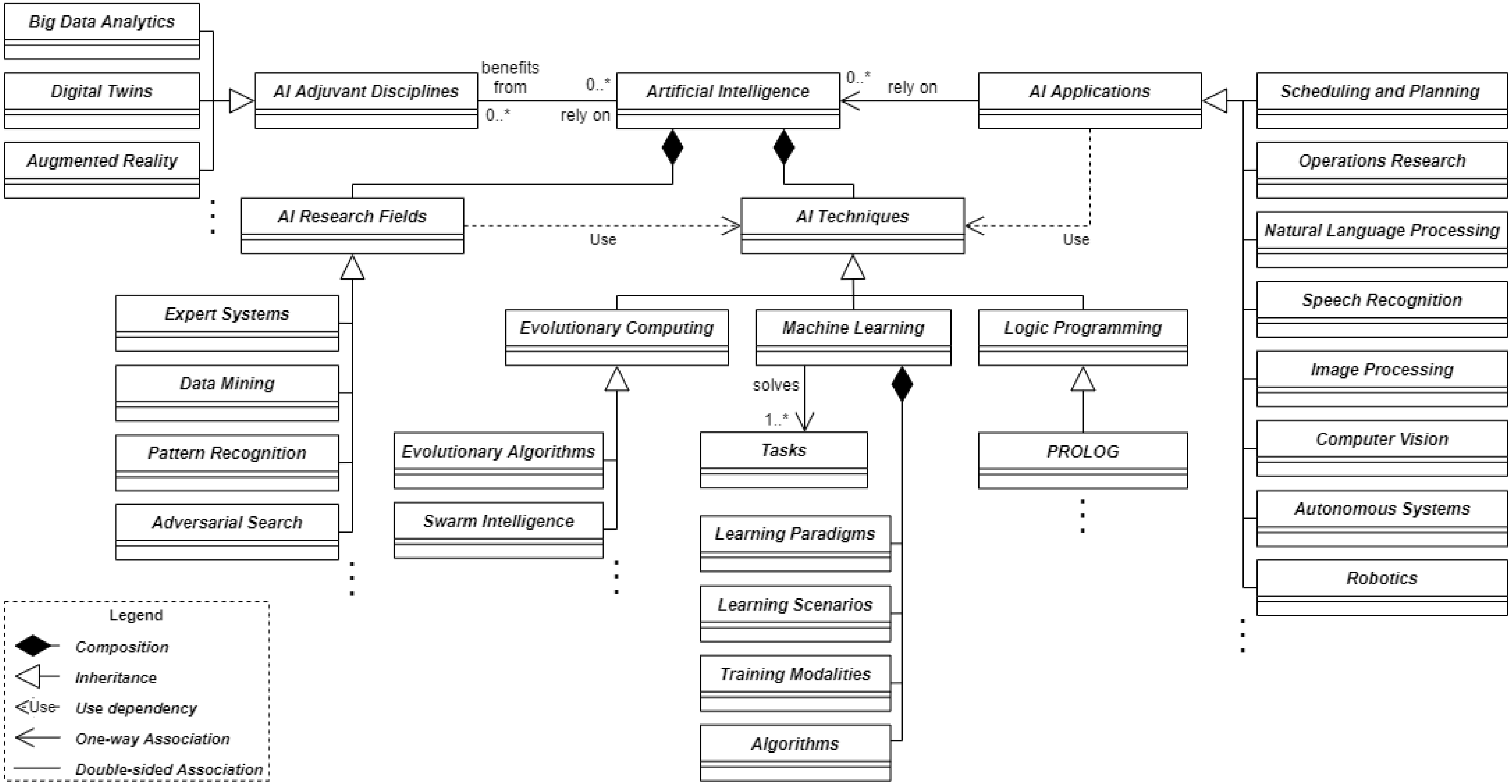


Image source: Wikimedia

What is artificial intelligence?

- Artificial Intelligence (AI) Any algorithm allowing a machine to **mimic a human behavior**
  - Pattern Recognition (PR) **Looking for pattern in the data** (e.g. correlations) to predict model the behavior of a system
    - Machine Learning (ML) Algorithms making a machine able **to learn from examples**
      - Artificial Neural Networks (ANN) A particular machine learning model based on the concept of **artificial neuron**
        - Deep Learning (DL) A **specific set of artificial neural networks** characterized by:
          - A huge number of artificial neurons, stacked in several layers
          - The ability of autonomously learn the best set of feature for a given task

A diagram based overview

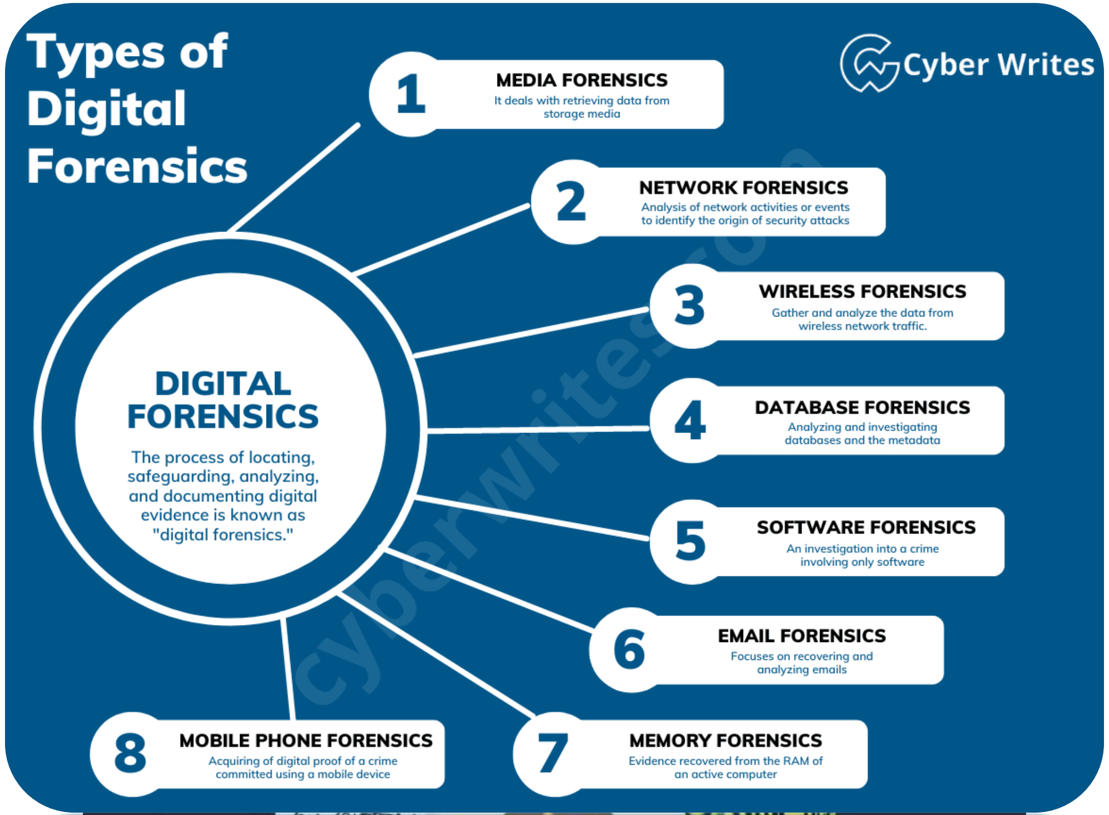


What does Forensics mean?

- Forensic scientists collect, preserve, and analyze evidence during the course of an investigation.

- The word forensic comes from the Latin word forensis: public, to the forum or public discussion; argumentative, rhetorical, belonging to debate or discussion. A relevant, modern definition of forensic is: relating to, used in, or suitable to a court of law. <u>Any science used for the purposes of the law is a forensic science.</u>

- Is it possibile to consider AI in these contexts? Would it be a ***friend*** or a ***foe***?



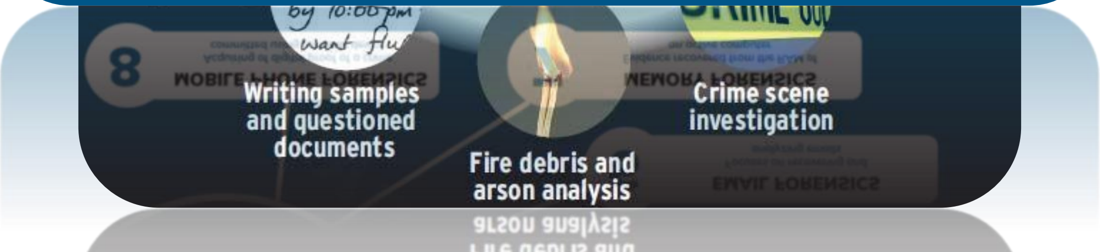


Image source: Wikimedia

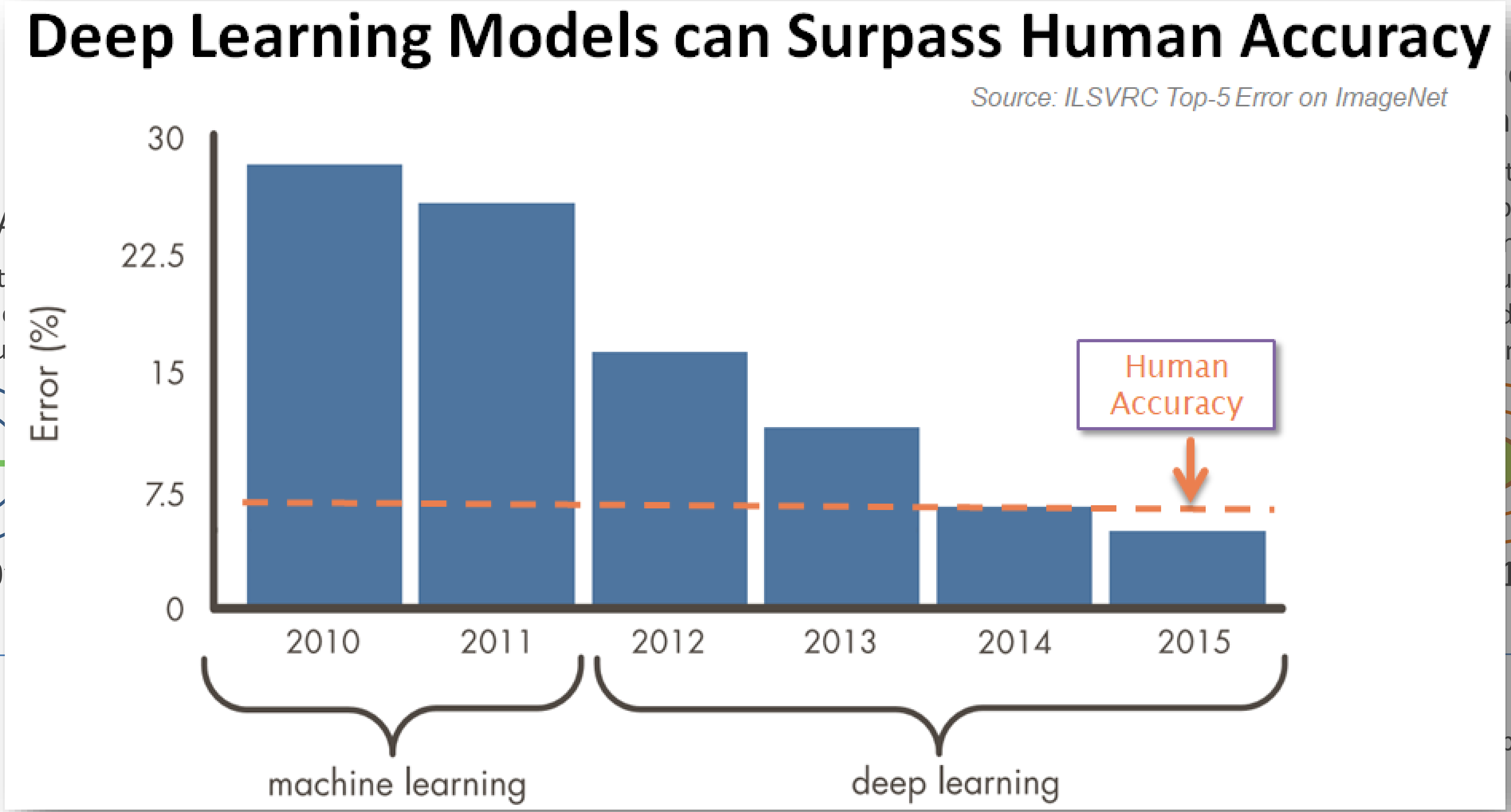
Background and context

Classical ML vs Deep Learning: AI Timeline

Early A

Researches start
the idea of a ma
to think like a hu

1940



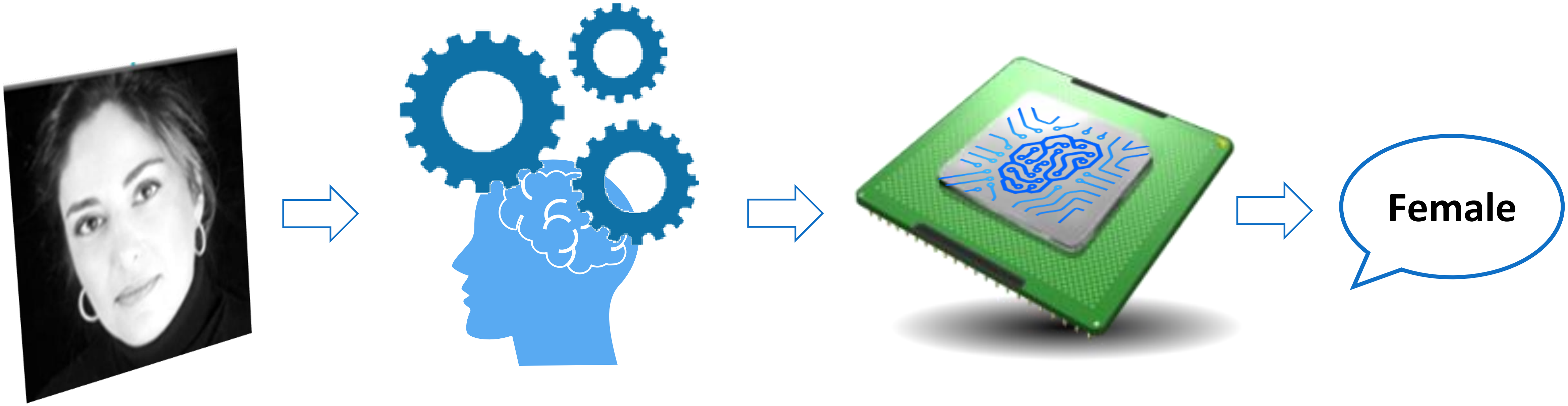
eep
g Era

tical solutions
bility of huge
ning data and a
utational
d deep learning
ne

12

Learning

Machine Learning



Input Data

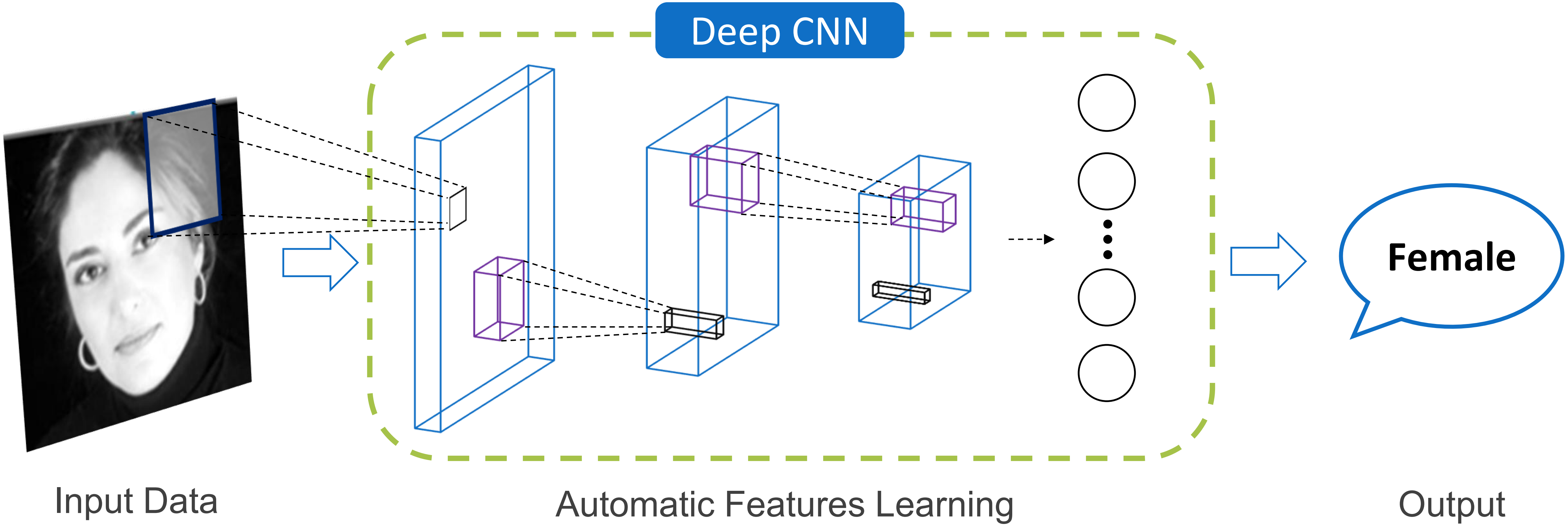
Domain Expert
(feature design)

Classifier
(es. Neural Network)

Output

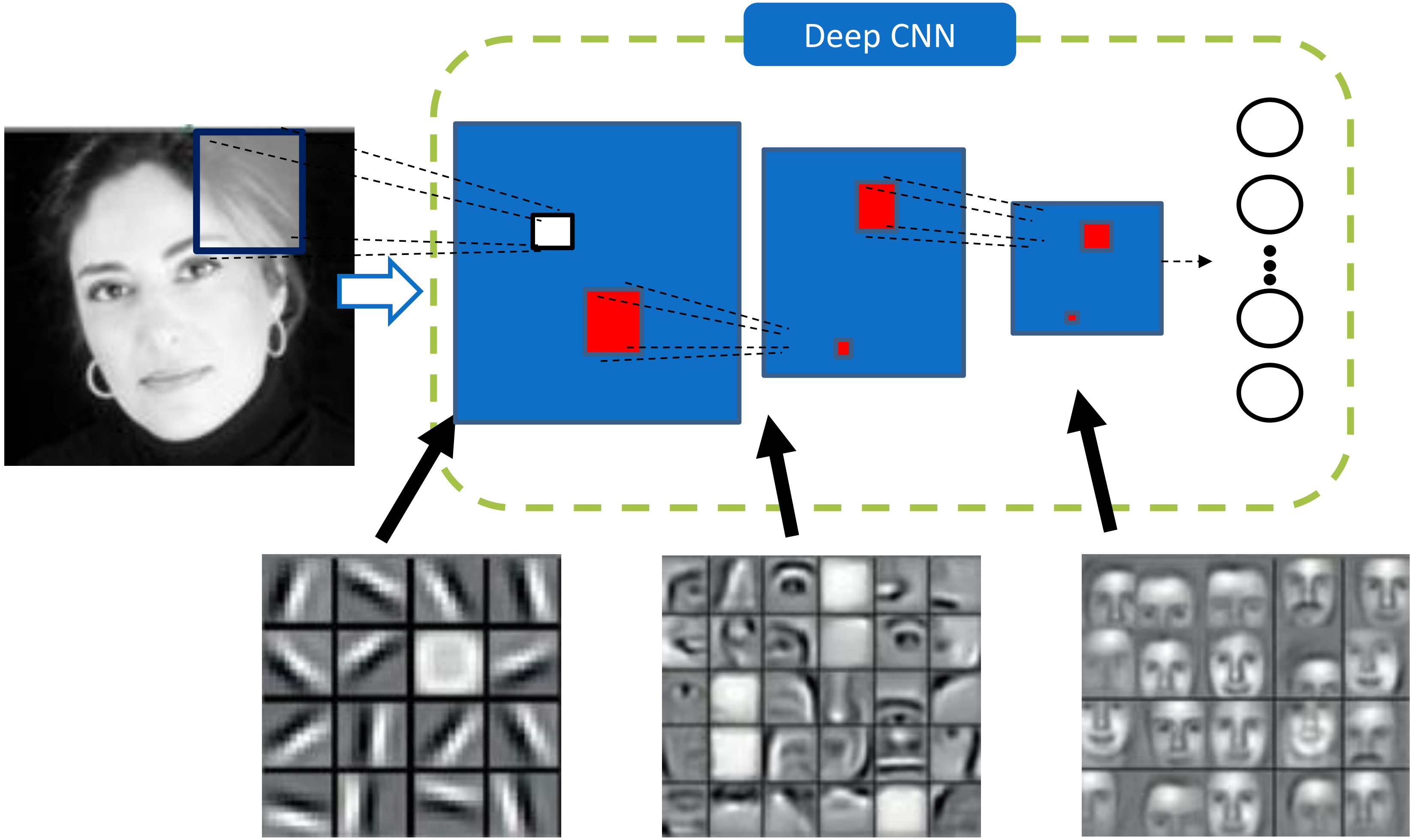
- The domain expert (a human) manually designs the desired set of features to extract

Deep Learning



✓ The model autonomously learn the most suitable features for the task

Deep Convolutional Neural Networks (CNNs)



- ✓ **Concepts hierarchy** (like human brain)
- ✓ **Autonomous features learning**
- ✓ **Generalization** ability
- ✓ **Domain adaptation**

The burden of CNN

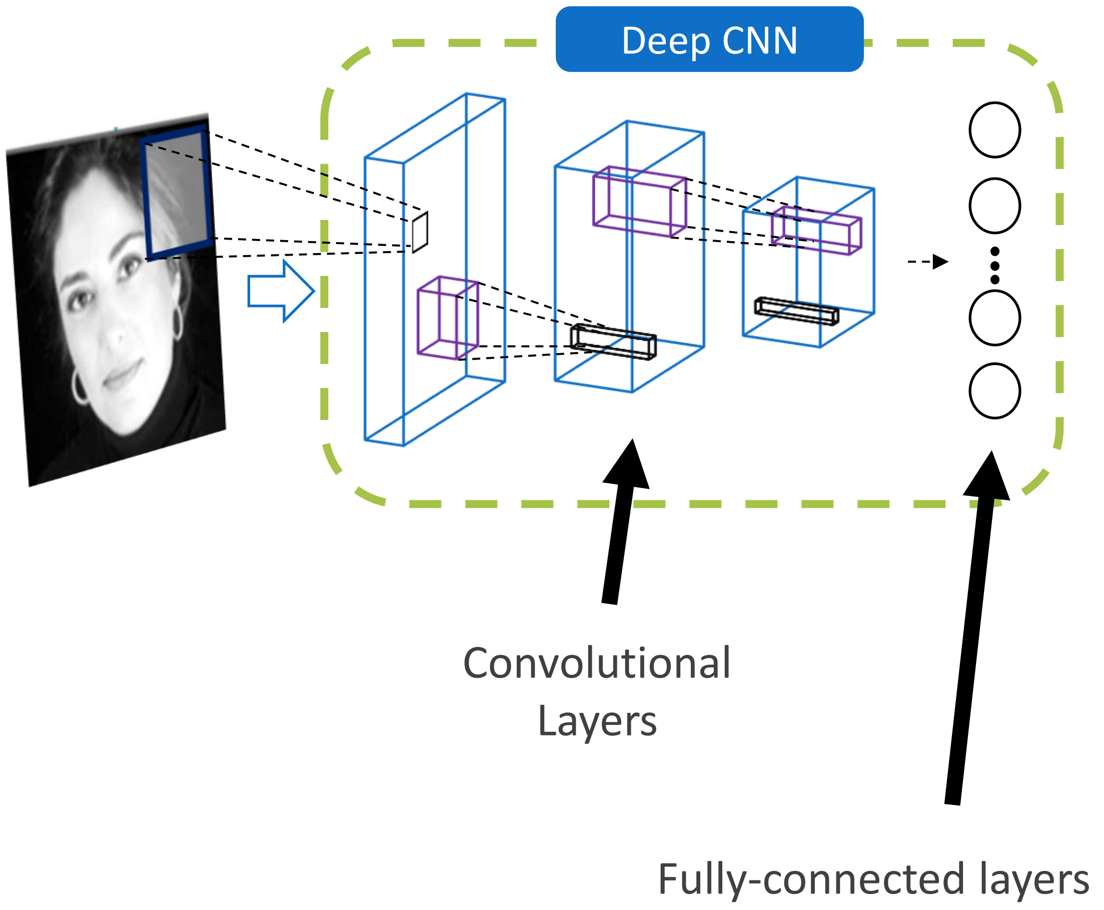
- The term deep refers to the number of layers and thus, indirectly, to the total number of neurons (parameters)
- To avoid incurring in over/under-fitting, a suitable number of annotated samples is required
  - Gathering a big dataset is a difficult, expensive and time-consuming task, that can become even harder in domains where a large number of samples is difficult to collect

Model	Top-1 Accuracy	Top-5 Accuracy	Parameters	Depth
AlexNet [95]	0.593	0.818	60,965,224	8
VGG16 [164]	0.715	0.901	138,357,544	23
VGG19 [164]	0.727	0.910	143,667,240	26
MobileNet [79]	0.665	0.871	4,253,864	88
DenseNet121 [80]	0.745	0.918	8,062,504	121
Xception [33]	0.790	0.945	22,910,480	126
InceptionV3 [178]	0.788	0.944	23,851,784	159
ResNet50 [76]	0.759	0.929	25,636,712	168
DenseNet169 [80]	0.759	0.928	14,307,880	169
DenseNet201 [80]	0.770	0.933	20,242,984	201
InceptionResNetV2 [176]	0.804	0.953	55,873,736	572

- Brief list of some famous CNNs designed to face the ImageNet [4] classification challenge. For each network, the top-1 and top-5 accuracy, together with the number of parameters and layers (depth) are reported [5].

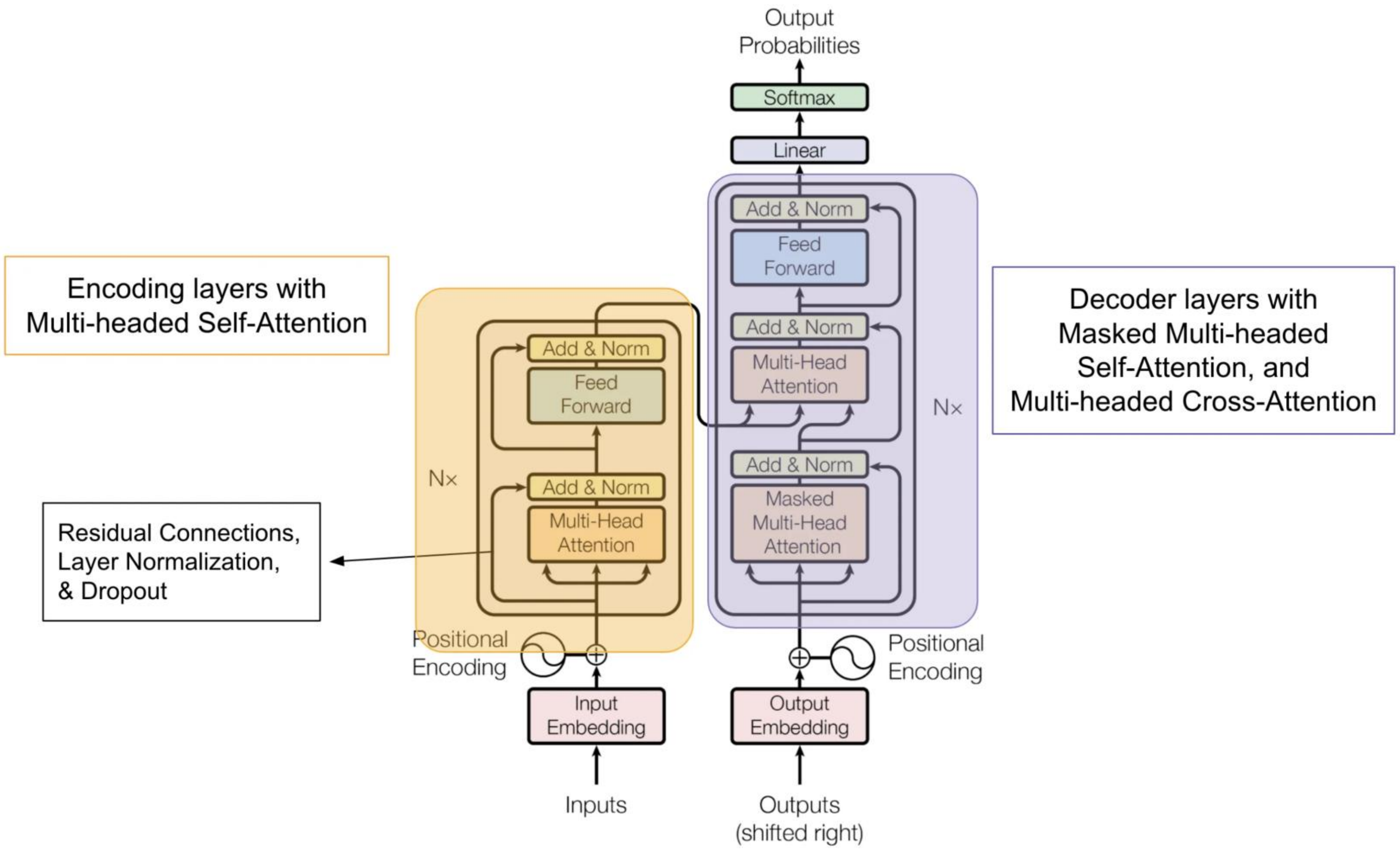
CNN knowledge transfer

- Deep CNNs consist of two part:
  - Convolutional Layers, learning how to extract the best set of features for the task under analysis
  - Fully Connected Layers, using the learnt features to perform the actual classification
- It has been demonstrated that CNNs pre-trained on ImageNet (a famous competition with 1.5M training samples over 1000 classes) generalize very well also on very different tasks
- One of the most used approach is fine-tuning, consisting in replacing the classification layer with one having as many neurons as the number of classes in the new task



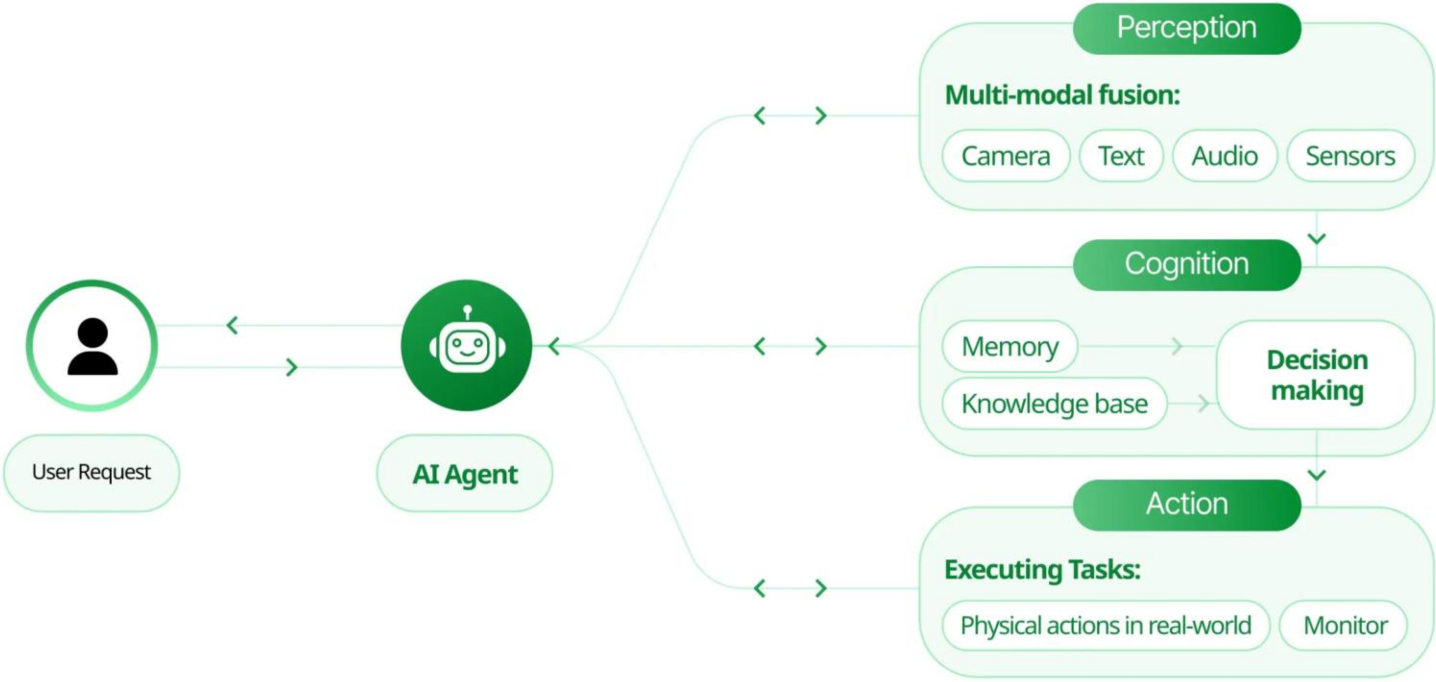
Recent advances: the transformer architecture

- An encoder-decoder architecture exploiting the attention mechanism to address the training data problem
- Connecting several transformer modules together results in (often very) big models able to be trained with “small” labelled datasets
- Are demonstrating interesting performance in natural language processing and reasoning



Recent advances: the agentic architecture

- An "updated" version of the classical agent-evironmnt schema, in which the agent is ML-based (often a LLM), orchestrating the whole set of opeations
- Not all the operations are ML-based. Instead, several features are supported through third-party tools
- In some scenarios, the systems consists of several AI-based agents, collaborating each other



So far so good? But…

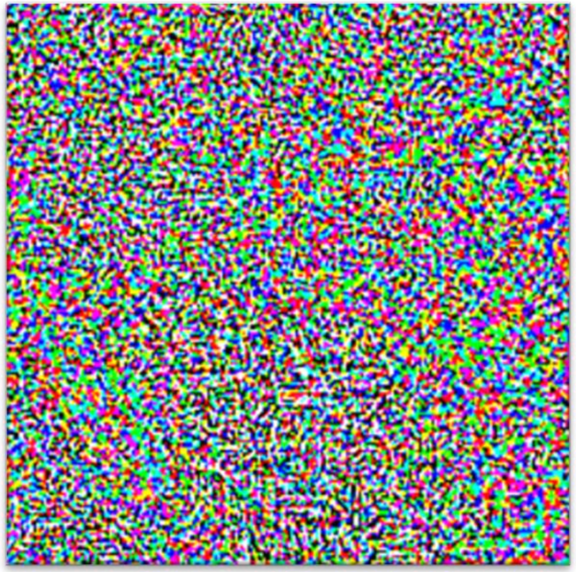
Deceivable AI

- Optical illusion is the term used to describe figures able to mislead the human visual perception system
- Since the hierarchical structure of deep neural networks is inspired by the human brain, is it possible to mislead them similar to the way optical illusions mislead us?

Adversarial Perturbations

- Injection of an imperceptible noise in order to mislead a CNN

Cat (Prob. 99,2%)



Noise

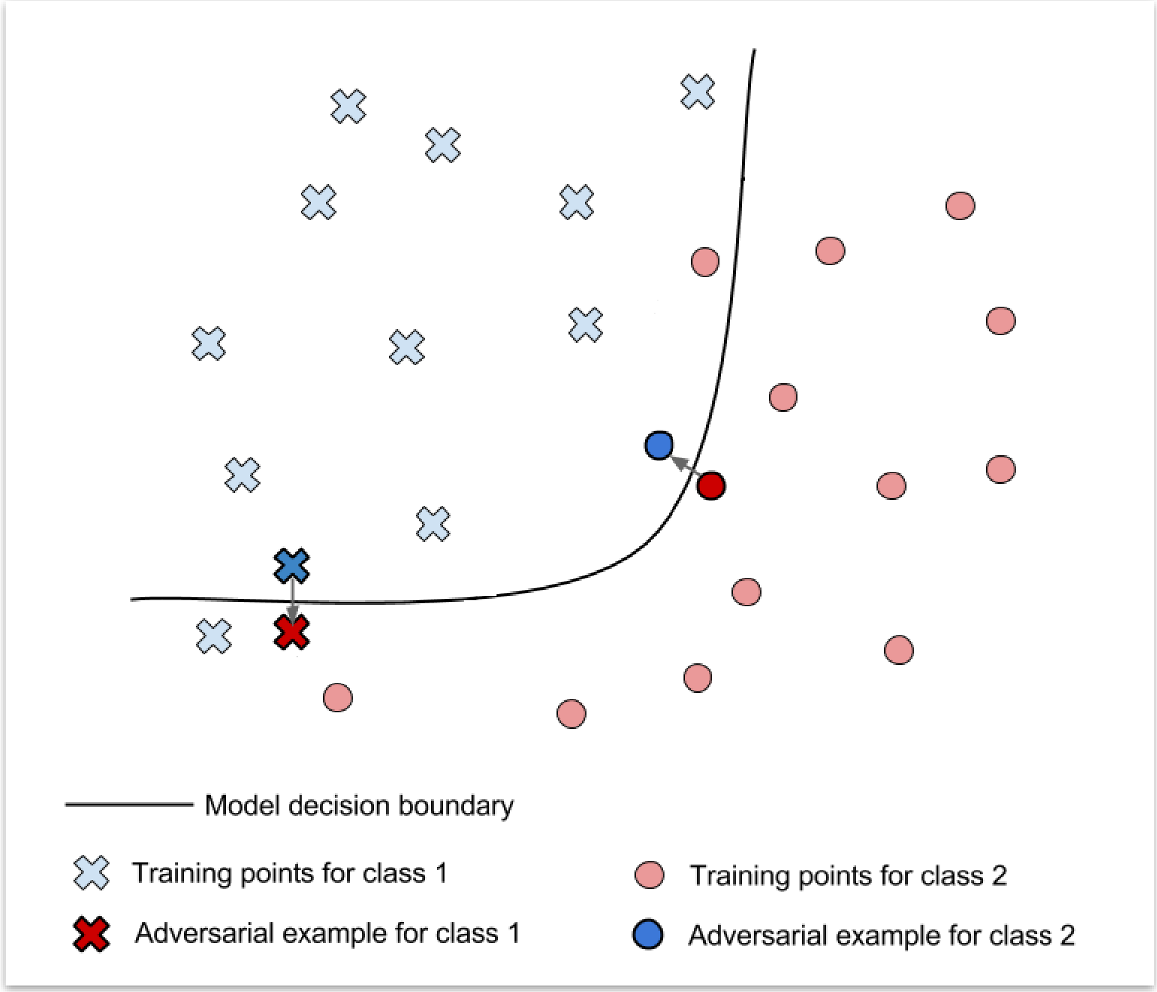
Dog (prob. 89,7%)

Szegedy et al. “Intriguing properties of neural networks”, arXiv:1312.6199 (2014)
Moosavi-Dezfooli, et al. “Deepfool: a simple and accurate method to fool deep neural networks”, in Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (2016)

The logic behind adversarial perturbations

The math behind adversarial perturbations

- The aim is to move the target sample beyond the model decision boundary
  - Gradient Methods exploits the gradients information with respect to the input in order to determine the best perturbation to add
  - Genetic Methods “randomly” change input data until a fitness function says that the obtained perturbation is able to mislead the classifier



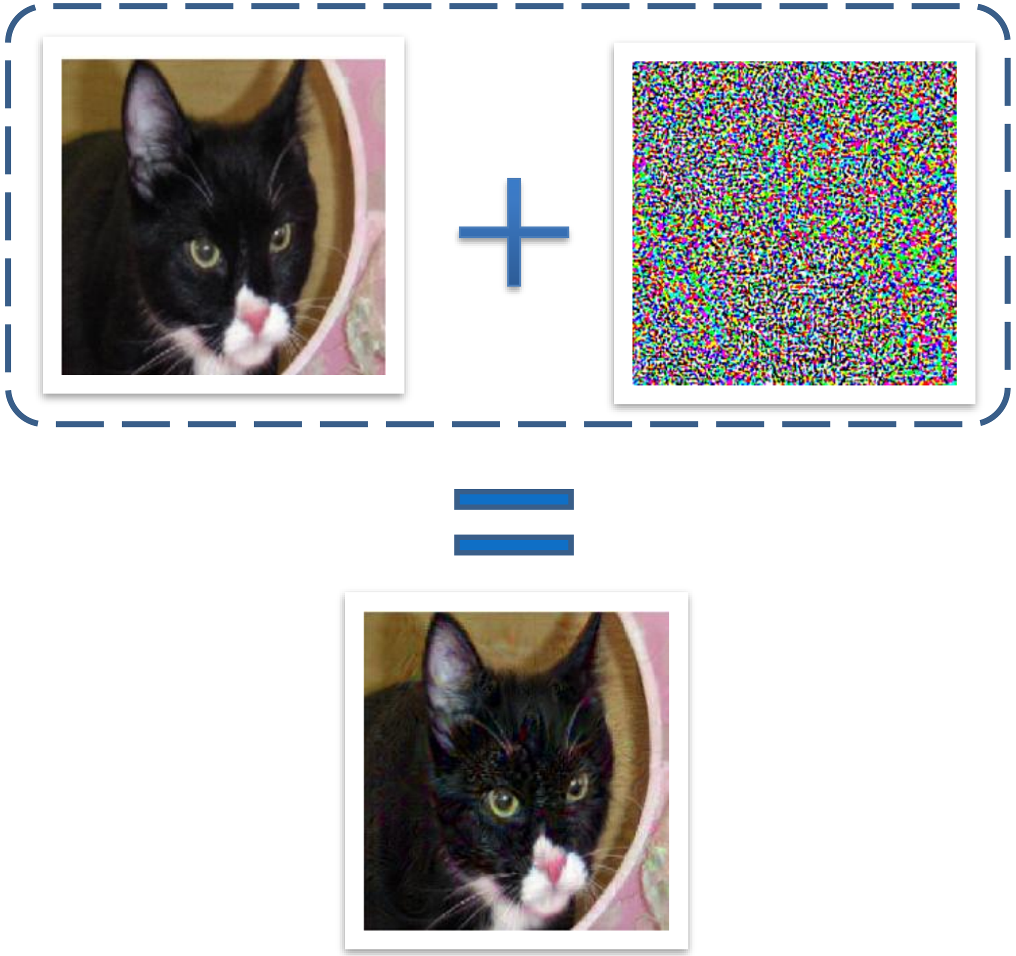
Exploiting adversarial perturbations

- The vulnerability of CNNs to adversarial perturbations could open new possibilities for privacy protection and several authors have already exploited adversarial perturbation to mislead CNNs trained human faces

- Among all, gender misleading is one of the most analyzed task[1,2,3]

- However, all the approaches so far proposed directly feed the perturbed image to the classifier

- The perturbations, some of which really sneaky, are designed to change a target image, and thus are not applicable in a real world scenario



[1] V. Mirjalili et al. "Semi-adversarial networks: Convolutional autoencoders for imparting privacy to face images“ in ICB (2018)
[2] A. Rozsa et al. "Facial attributes: Accuracy and adversarial robustness“ in Pattern Recognition Letters (2017)
[3] A. Othman et al. "Privacy of facial soft biometrics: Suppressing gender but retaining identity“ in ECCV (2014)

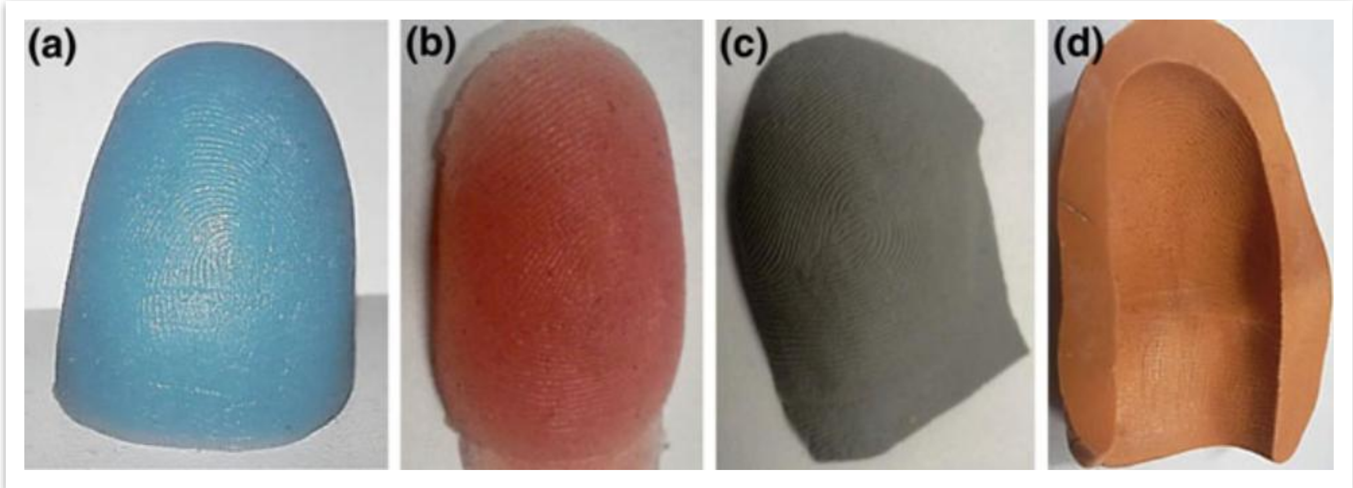
Exploiting Adversarial AI

Biometric based Authentication Systems

- Biometric Based Authentication Systems (BASs) allows to recognize a subject according to “what they are” rather then on “what they use”
  - Among all, fingerprints are the most used (Fingerprint based Authentication Systems – FAS)
  - As for other domains, CNN has proved to be very effective in analyzing biometrics data
- Given CNNs blindspots, might they make BAS less robust against malicious agents?

Presentation attack

- As for any other authentication means, it is possible to attack a BAS by using a counterfeit replica of the target subject biometrics
- Since the attack consists in “presenting” the fake replica to the scanner, this type of attacks goes under the name of *Presentation Attack*

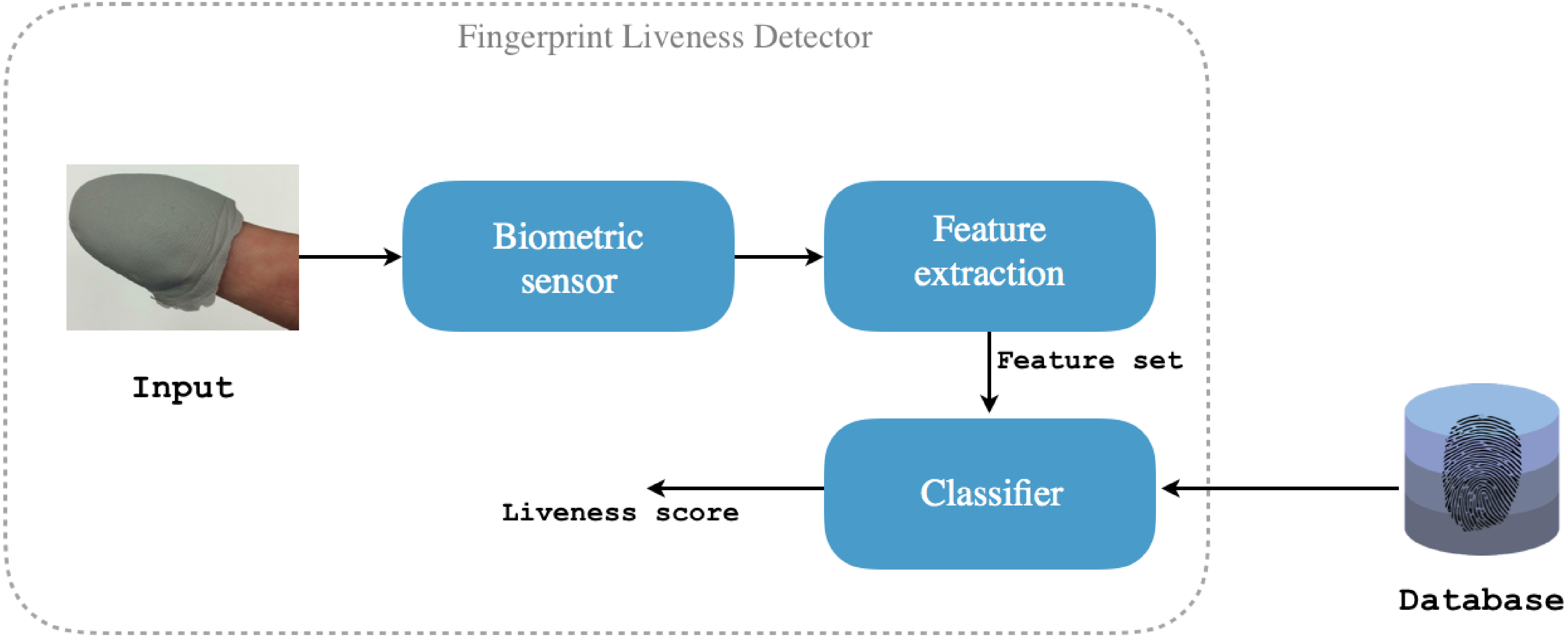


- Artificial finger replicas made using GLS (a), Ecoflex (b), Liquid Ecoflex (c) and Modasil (d)

Fingerprint Presentation Attack

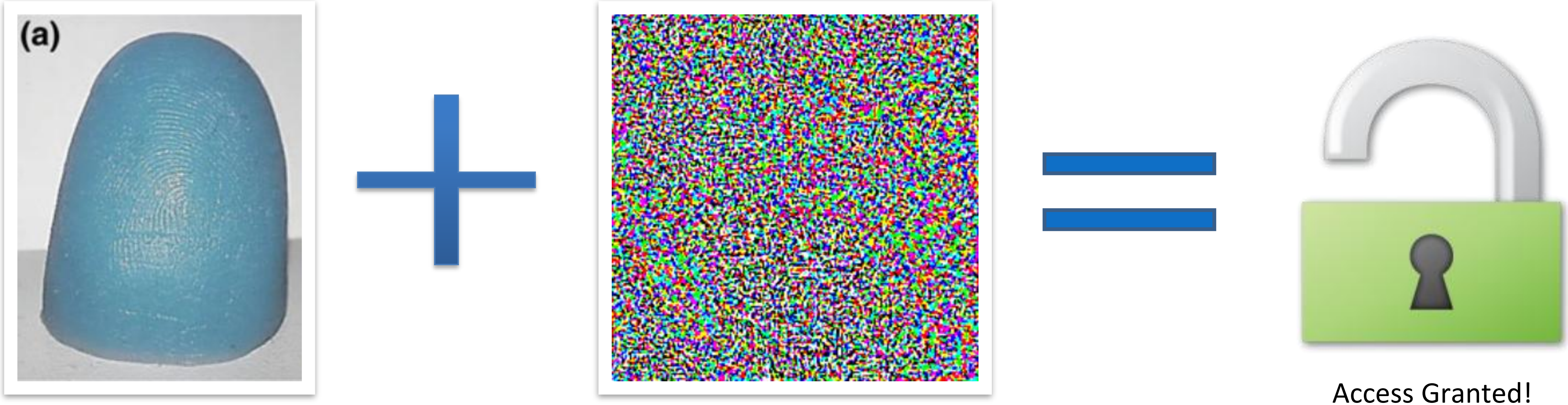
- Present artificial replicas of fingerprints to a sensor
- **Different materials** such as silicone, gelatine, play-doh, ecoflex, 2D printed paper, 3D printed material, latex, etc.
- **Consensual method**: collaborative user, acquisition with cast of the finger
- **Un-consensual method**: acquisition from latent fingerprints

Fingerprint Presentation Attack Detection (FPAD)



Adversarial presentation attack

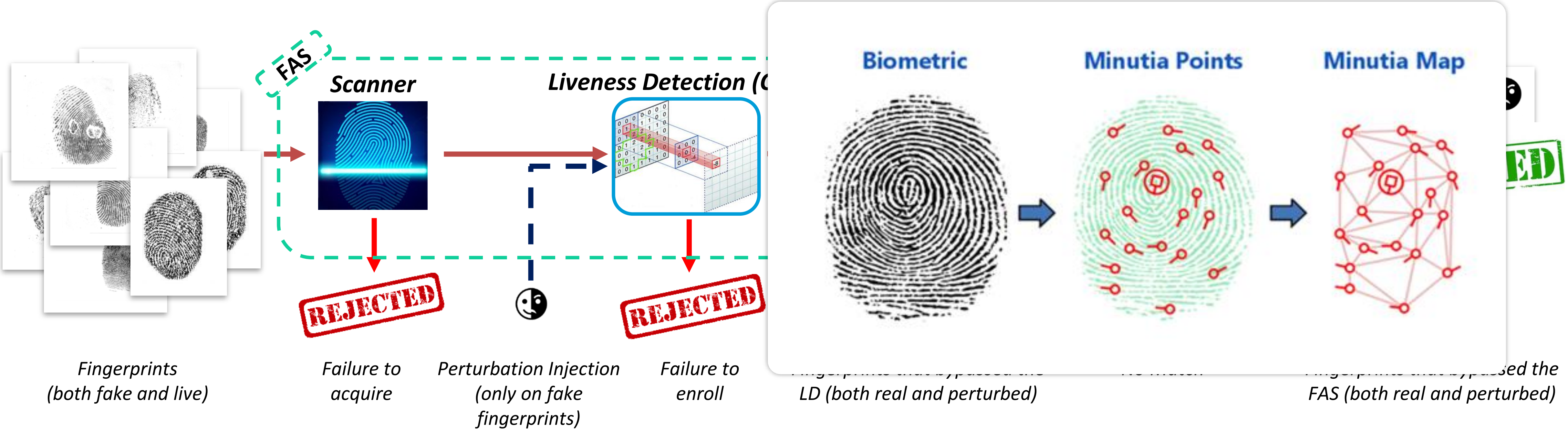
- Can we exploit adversarial perturbation to alter a fake fingerprint acquisition such that it mislead a CNN liveness detector?



Access Granted!

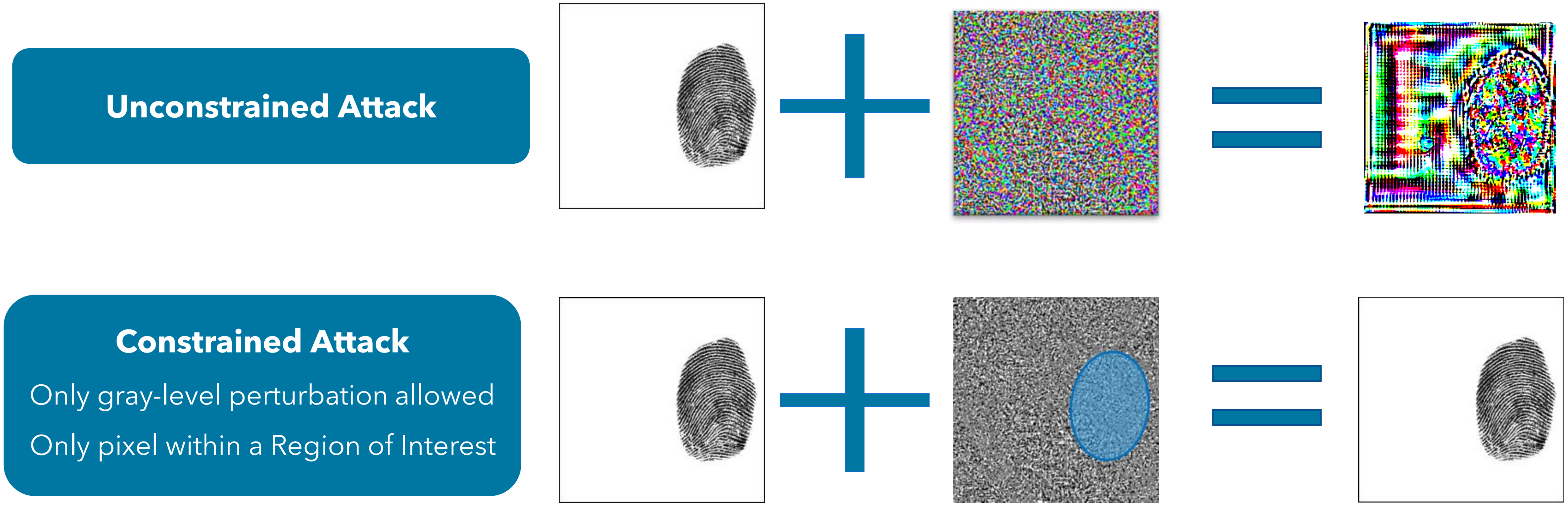
Adversarial presentation attack

- The aim is to understand if and to what extent adversarial perturbation can affect a FAS , tying to answer to:
  - how vulnerable are CNN based liveness detectors to adversarial perturbation?
  - Is the counterfeit footprint still recognized by the authentication system or had its main characteristics been destroyed by the perturbation attack?



A constrained attack

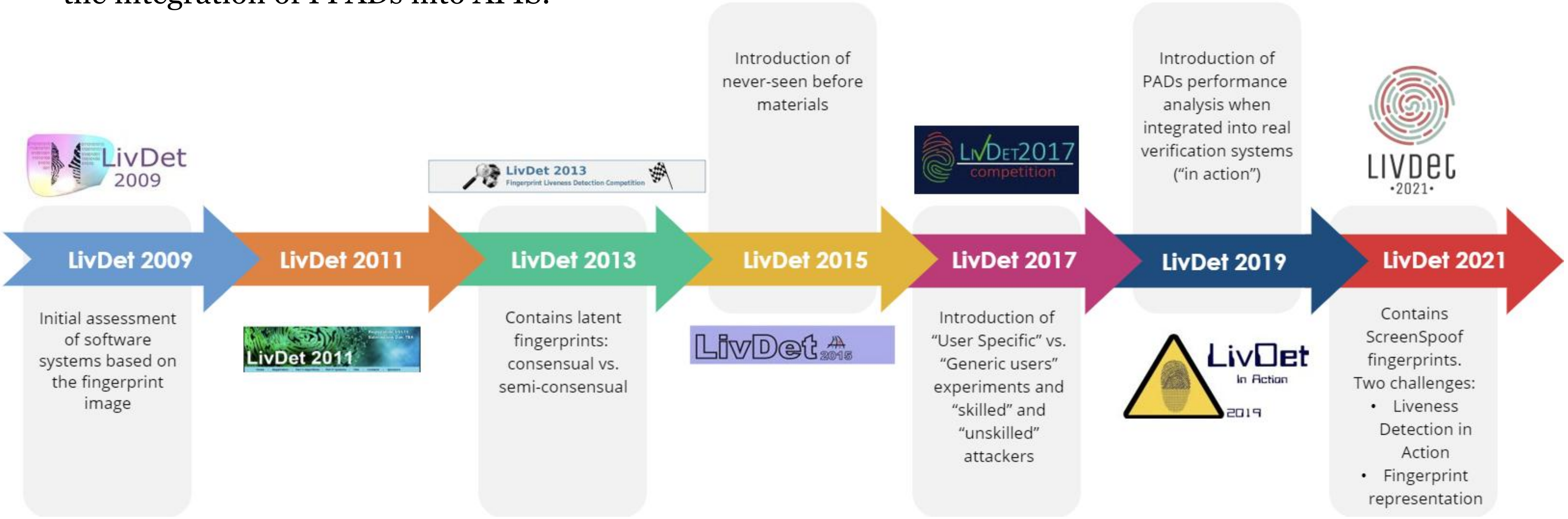
- Fingerprints images are different from natural images and the injected noise could be very visible and difficult to hide



Fingerprint Liveness Detection Competition

- The Liveness Detection (LivDet) Competition is a biennial competition in which participants are challenged to identify spoofs from live samples

- Each edition has its own distinctive set of challenges that competitors must overcome:
  - ✓ the presence of different materials for the training and test sets
  - ✓ the integration of FPADs into AFIS.



The two sides of adversarial attacks in FPAD

- FPAD has seen the rise and establishment of Convolutional Neural Networks (CNNs) as an effective approach to the problem:
  - ✓ CNNs obtain state-of-the-art performance in identifying a large number of fake samples

- ALD represents a deep learning-based fingerprint liveness detection whose aim is to exploit the experience matured as attackers to design an ad-hoc adversarial data augmentation strategy intended to increase the effectiveness of CNN-based presentation attack detection

😈 + 😇 = ALD

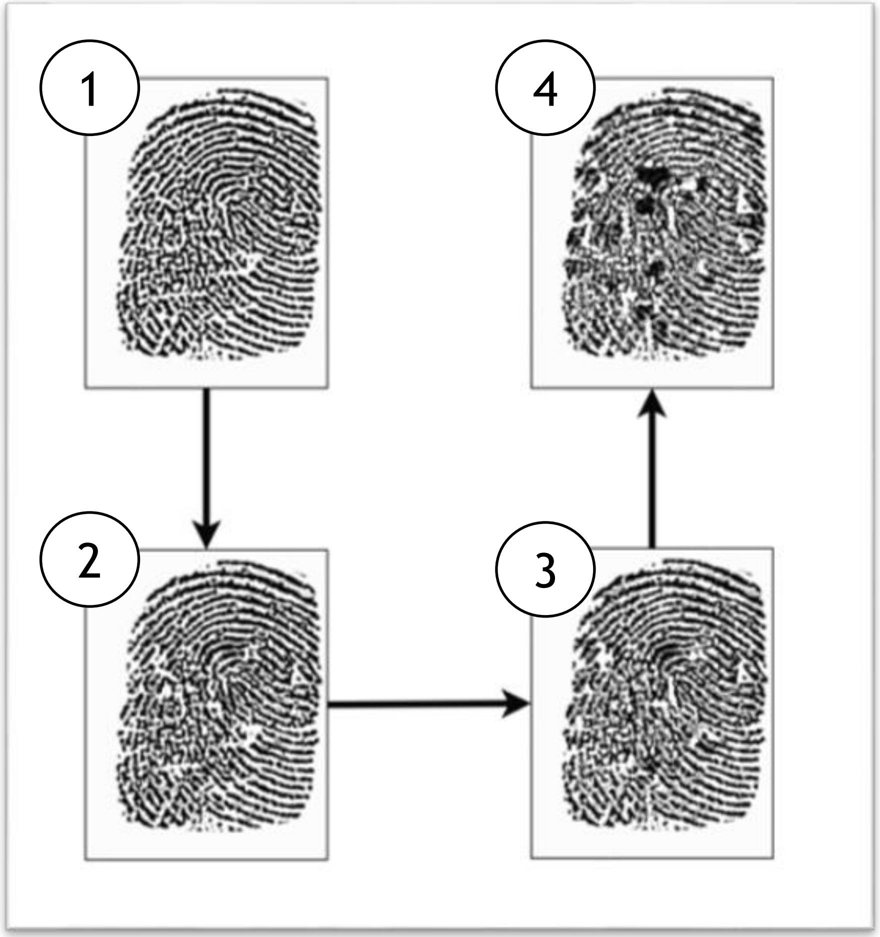
ALD has been submitted to the LivDet 2021 competition and obtained the first place out of 23 participants in the “Liveness Detection in Action track”.

Effects of Adversarial Samples for Training

- The idea is to realize an iterative training procedure consisting of three main steps:
  - Train a CNN-based Liveness Detector (LD) on the available data
  - Generate adversarial fingerprints (DeepFool) using the trained LD as target CNN
  - Add these new fingerprints to the pool of training data

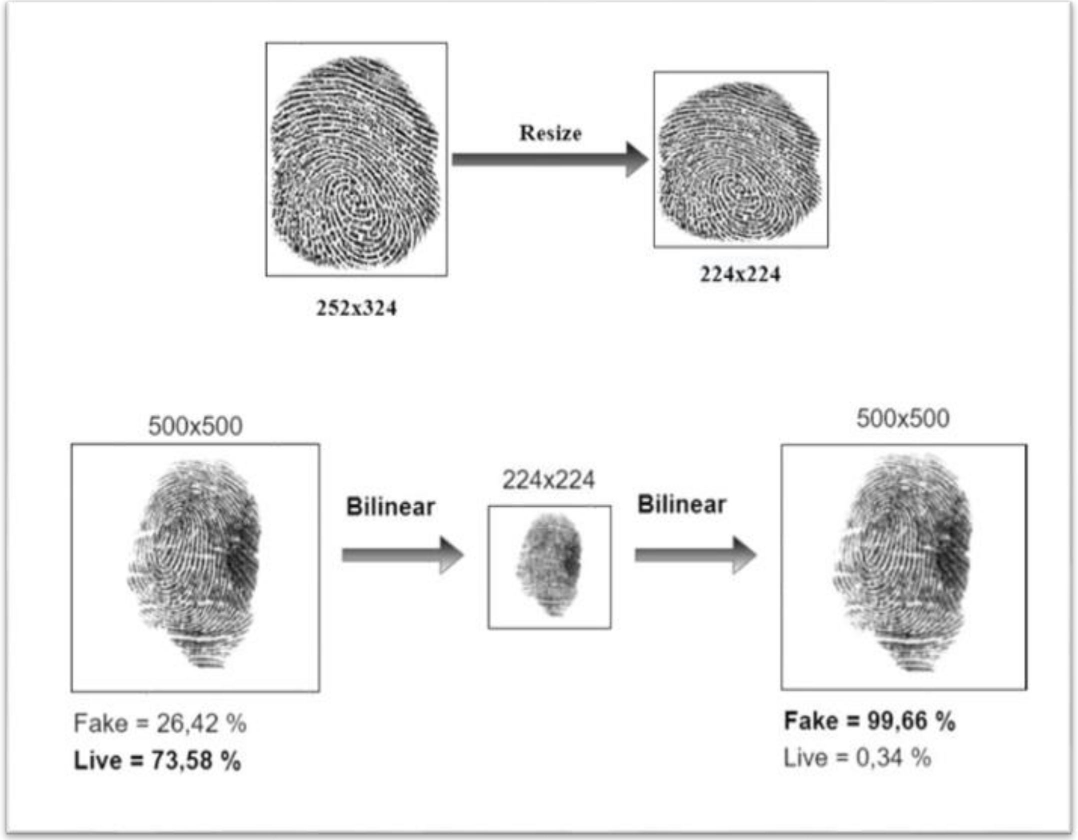
- Despite this "adversarial data augmentation" schema sounds legit, fingerprints are different from natural images:
  - It turned out that the adversarial attack success rate quickly decreases as the described training iteration increases
  - As a consequence, only the first iteration allows the adversarially trained CNN to increase its generalization ability

- The reason is that the so trained LD tends to be very robust against the attack, causing the adversarial algorithm to "destroy" the fingerprints



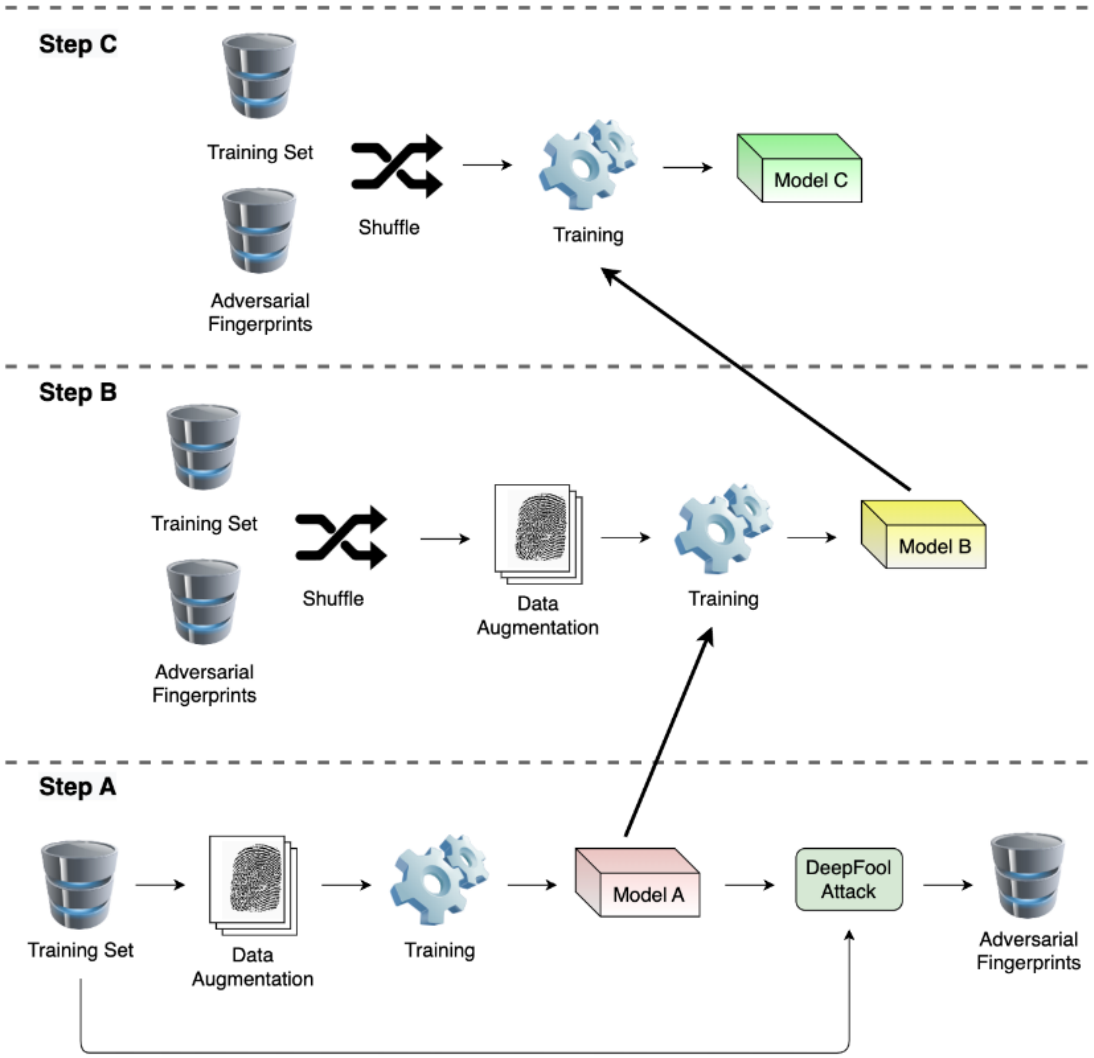
Effects of Image Resize

- Using a CNN also makes harder to deal with the size of the acquired fingerprint:
  - Different scanners acquire fingerprint at a different resolution, not necessarily resulting in “square” images
  - CNNs (almost always) expect a square image as input, having sizes (commonly) smaller than the acquired fingerprint
  - Image resize is not an option, as it tends to destroy details in the acquired fingerprint



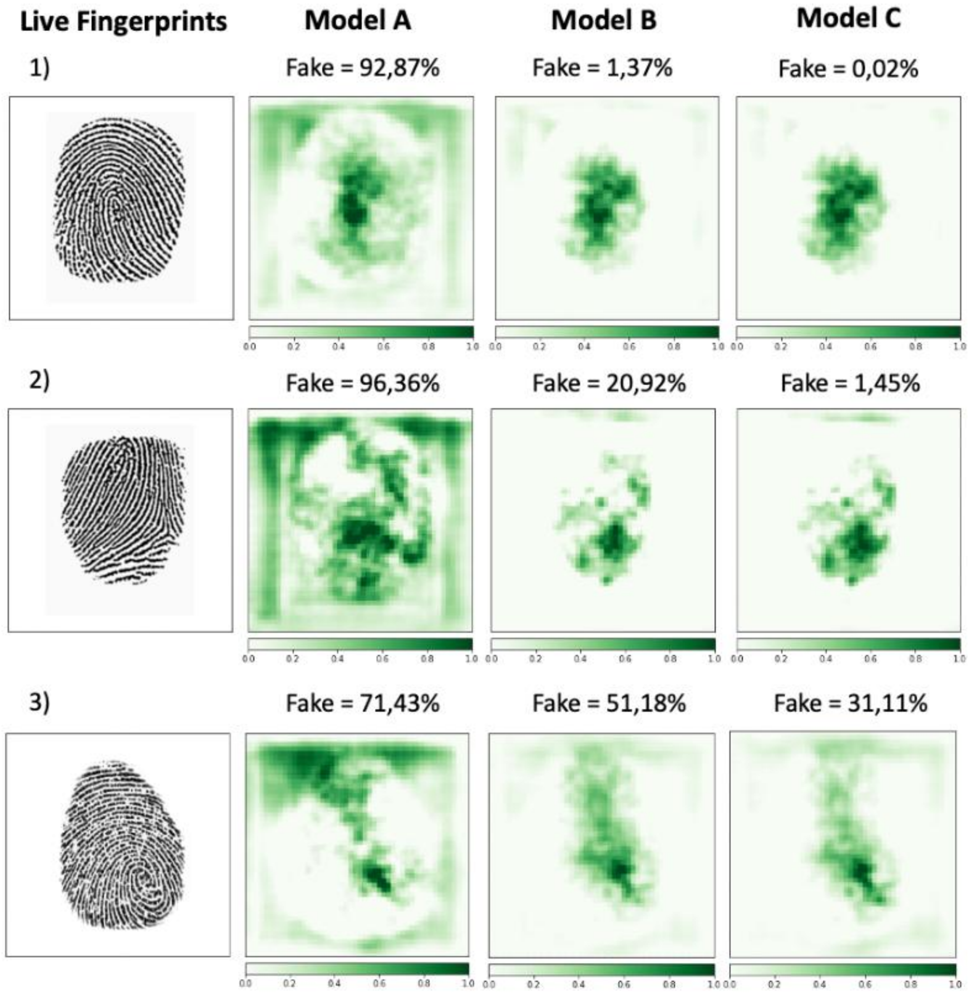
Resulting Training Schema

- The result is a three-steps training schema, where each step fine-tunes the model obtained in the previous one:
  - In the first step (A), the model (pre-trained on ImageNet) is fine-tuned on the challenge data. Data augmentation is used, limited to algorithms operating only on pixels intensity values (saturation, shading, etc.). Once the model is trained, the DeepFool algorithm is used to create a new dataset of adversarially perturbed fingerprint
  - In the second step (B), the model is further fine-tuned by using the new dataset consisting of both original and perturbed fingerprints. The same set of data augmentation algorithms has been used
  - In the third step (C), the model is fine-tuned for the last time by using the new dataset consisting of both original and perturbed fingerprints



Explainability Analysis

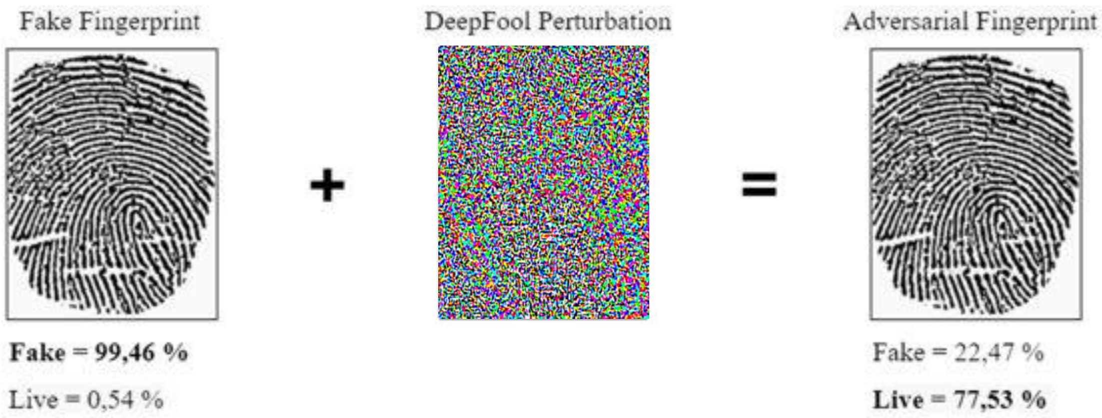
- Finally, with the aim of further analyzing the effects of the proposed approach, we performed an explainability analysis for each of the three steps
- The Occlusion method is used to highlight the most portion of the image that causes the liveness detector to classify the fingerprint as fake
- It can be noted that using adversarial fingerprints (models B and C) causes the model to mostly focus on the fingerprint inner region, almost ignoring borders and background



The two sides of adversarial attacks in FPAD

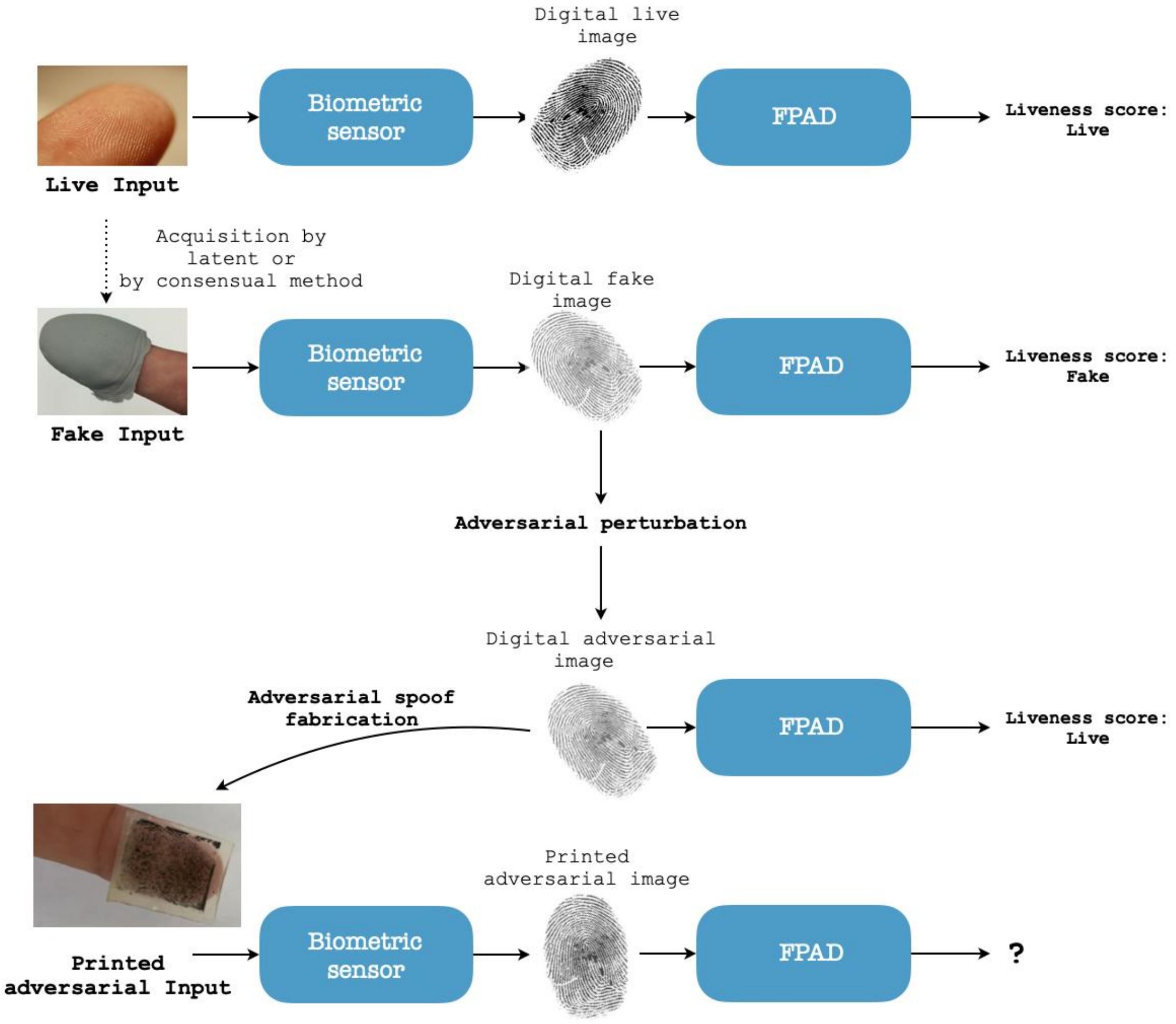
- The development of counter-anti-spoofing techniques:
  - ✓ approaches aimed at bypassing an AFIS despite being protected by an FPAD

- **Adversarial fingerprints**:
  - ✓ set of attacks designed to circumvent an FPAD by exploiting adversarial perturbations
  - ✓ Adversarial perturbation: a set of algorithms designed to mislead a target CNN by means of a specifically crafted noise



Fingerprint Adversarial Presentation Attack in the Physical Domain

- move the adversarial attacks from the digital domain to the physical one



Spoofs creation and acquisition

1. We create a positive mould by inverting the digital adversarial images
2. We printed several inverted fingerprints on the same sheet with a laser printer
3. A layer of latex is deposited over the prints of the individual perturbed fingerprints
4. We acquire each fake through the fingerprint sensor

Experimental Protocol

Dataset: LivDet 2015[1] -(Digital Persona – Latex)

FPAD: LivDet 2015 edition winner[2]

Adversarial perturbation algorithm: DeepFool[3]

Scanner	Image Size (px)	Live	Body Double	Ecoflex	Gelatine	Latex	Liquid Ecoflex	OOMOO	Playdoh	RTV	Woodglue
Biometrika	1000x1000	1000	-	250	250	250	250	-	-	250	250
CrossMatch	640x480	1500	300	270	300	-	-	297	281	-	-
DigitalPersona	252x324	1000	-	250	250	$\boxed{250}$	250	-	-	250	250
GreenBit	500x500	1000	-	250	250	250	250	-	-	250	250

[1] Mura, V., Ghiani, L., Marcialis, G.L., Roli, F., Yambay, D.A., Schuckers, S.A.: Livdet 2015 fingerprint liveness detection competition 2015. In: Biometrics Theory, Applications and Systems (BTAS), 2015 IEEE 7th International Conference on.pp. 1–6. IEEE (2015)

[2] Nogueira, R.F., de Alencar Lotufo, R., Machado, R.C.: Fingerprint liveness detection using convolutional neural networks. IEEE transactions on information forensics and security, 1206–1213 (2016)

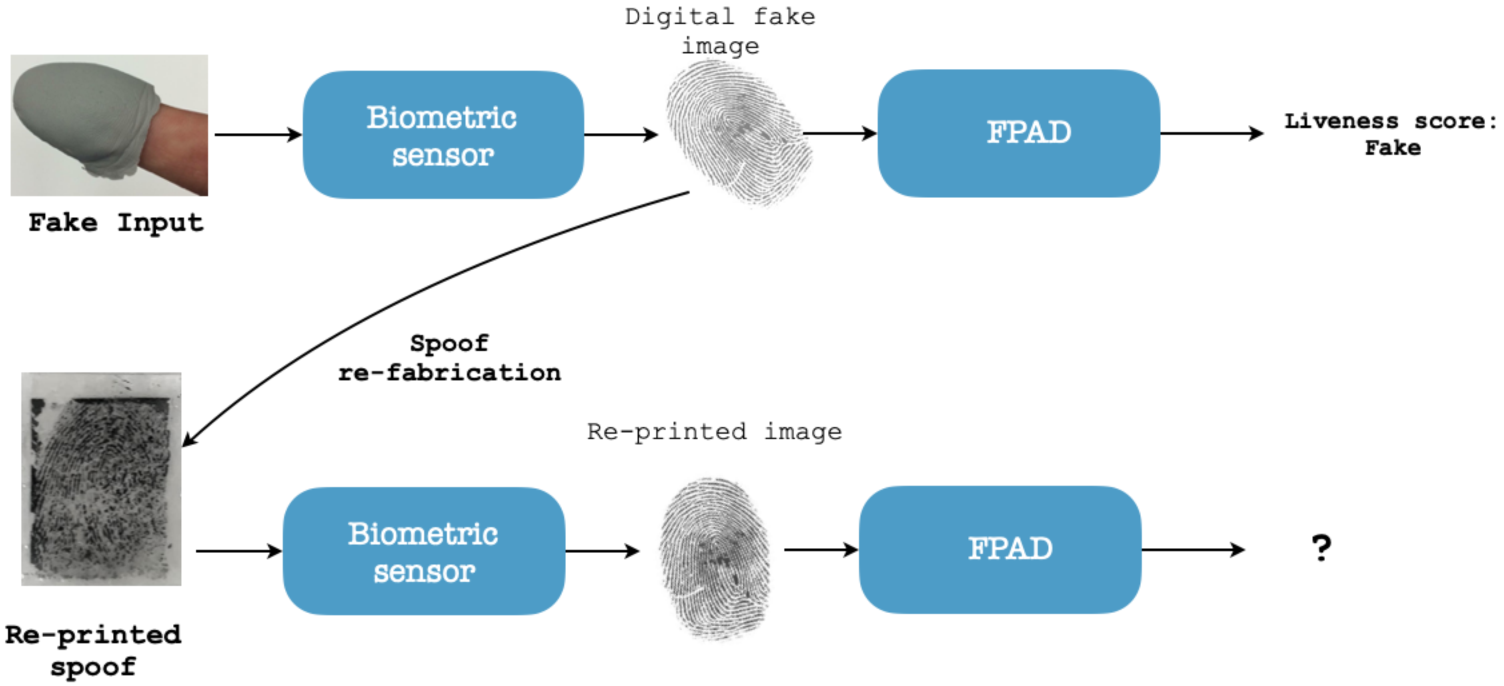
[3] Moosavi-Dezfooli, et al. "Deepfool: a simple and accurate method to fool deep neural networks", in Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (2016)

Experimental Protocol

- only fake fingerprint correctly classified as fake by the FPAD underwent the adversarial perturbation process (242 of 250)
- each spoof was acquired 10 with small rotations of the spoof on the sensor

Impact spoof re-fabrication

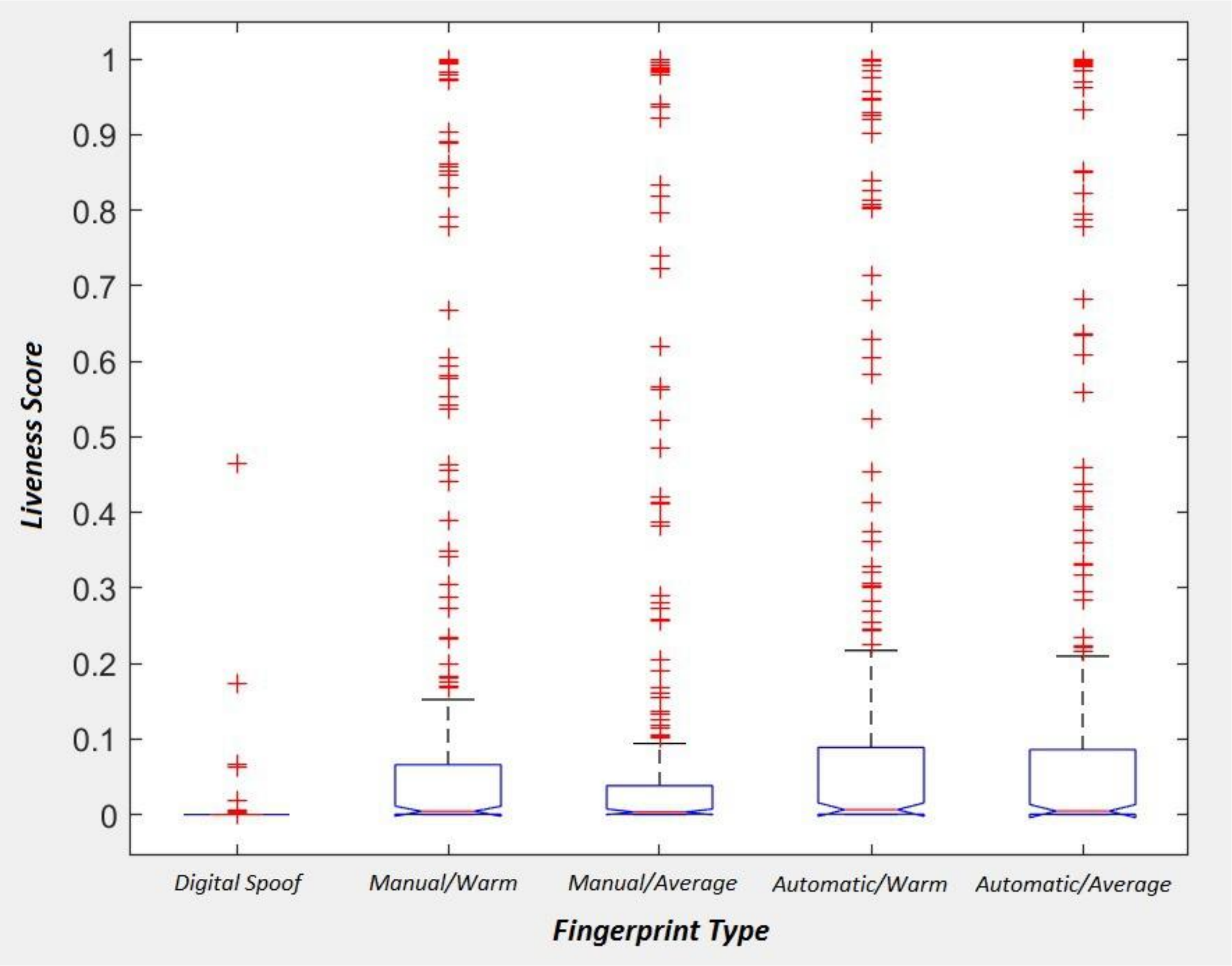
- verify how much the acquisition conditions and the pre-printing pre-processing influenced the liveness score



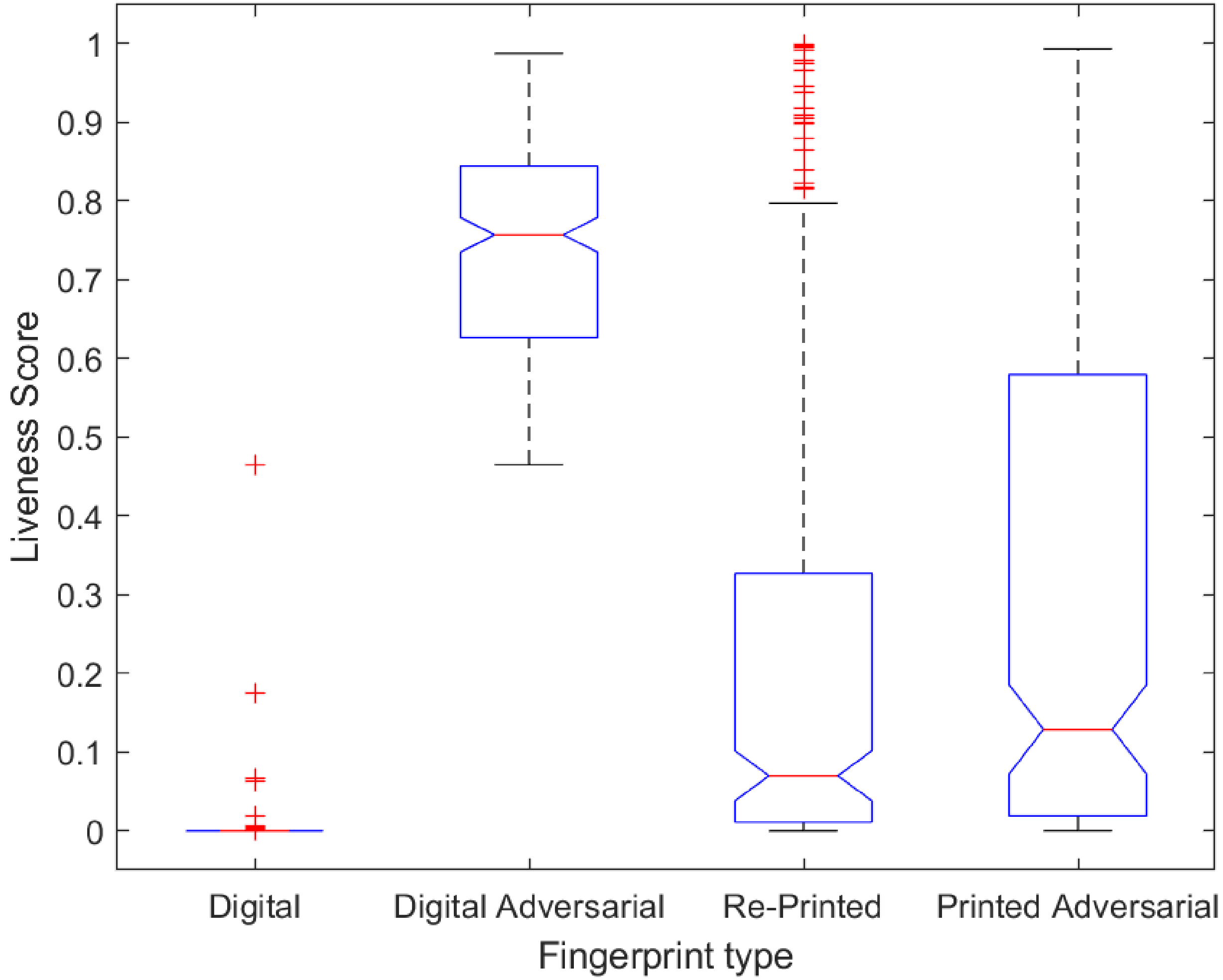
Acquisition conditions and pre-printing pre-processing influence

- Warm: T> 30° C
- Average: about T=20° C

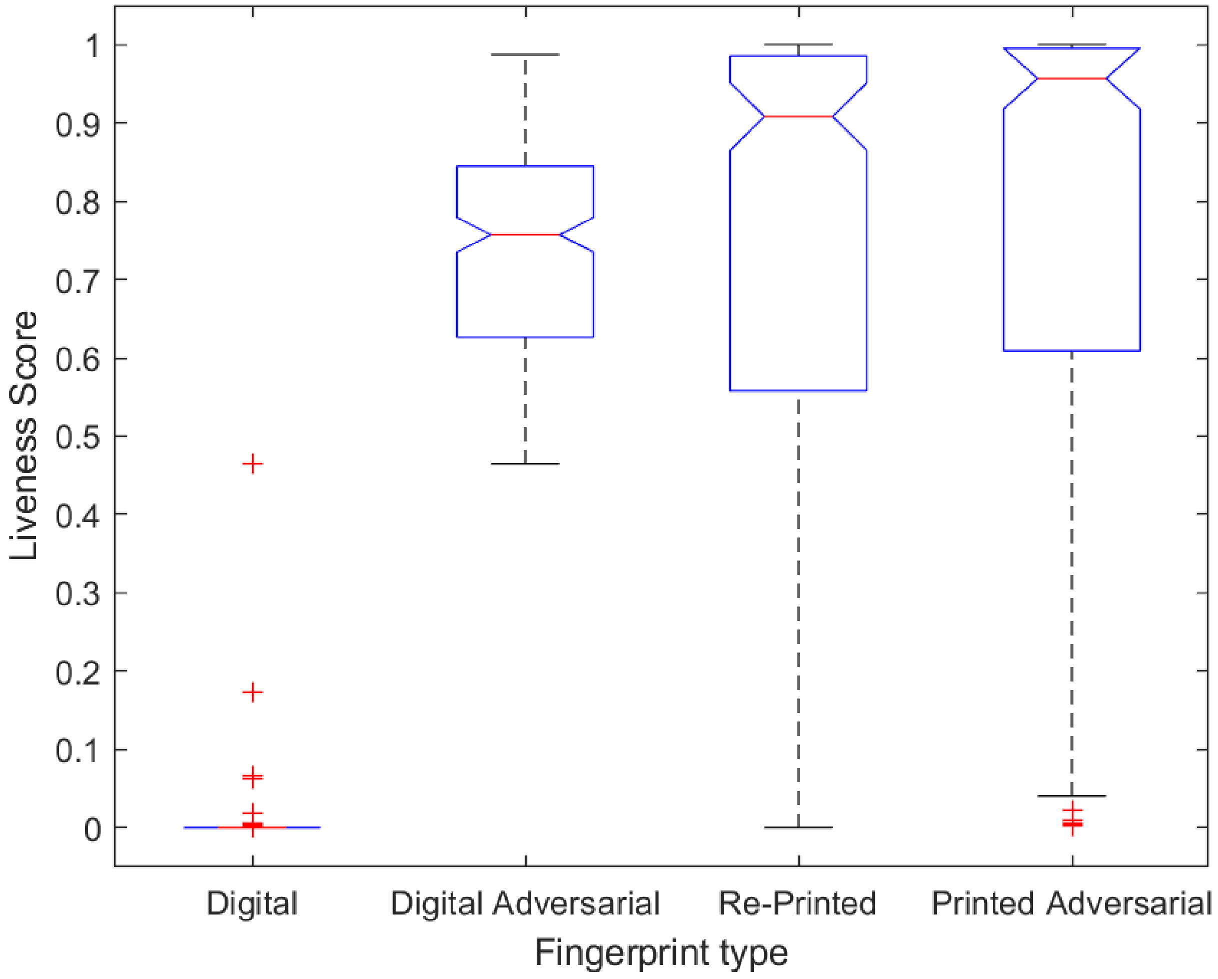
- Manual: inverting and resizing the fakes individually using an image editor
- Automatic: reversing and resizing the images via a MATLAB code



Results



Liveness Score
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
Digital
Digital Adversarial
Re-Printed
Printed Adversarial
Fingerprint type



Liveness Score
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
Digital
Digital Adversarial
Re-Printed
Printed Adversarial
Fingerprint type

AI Video Forensics

Discriminative vs Generative Models

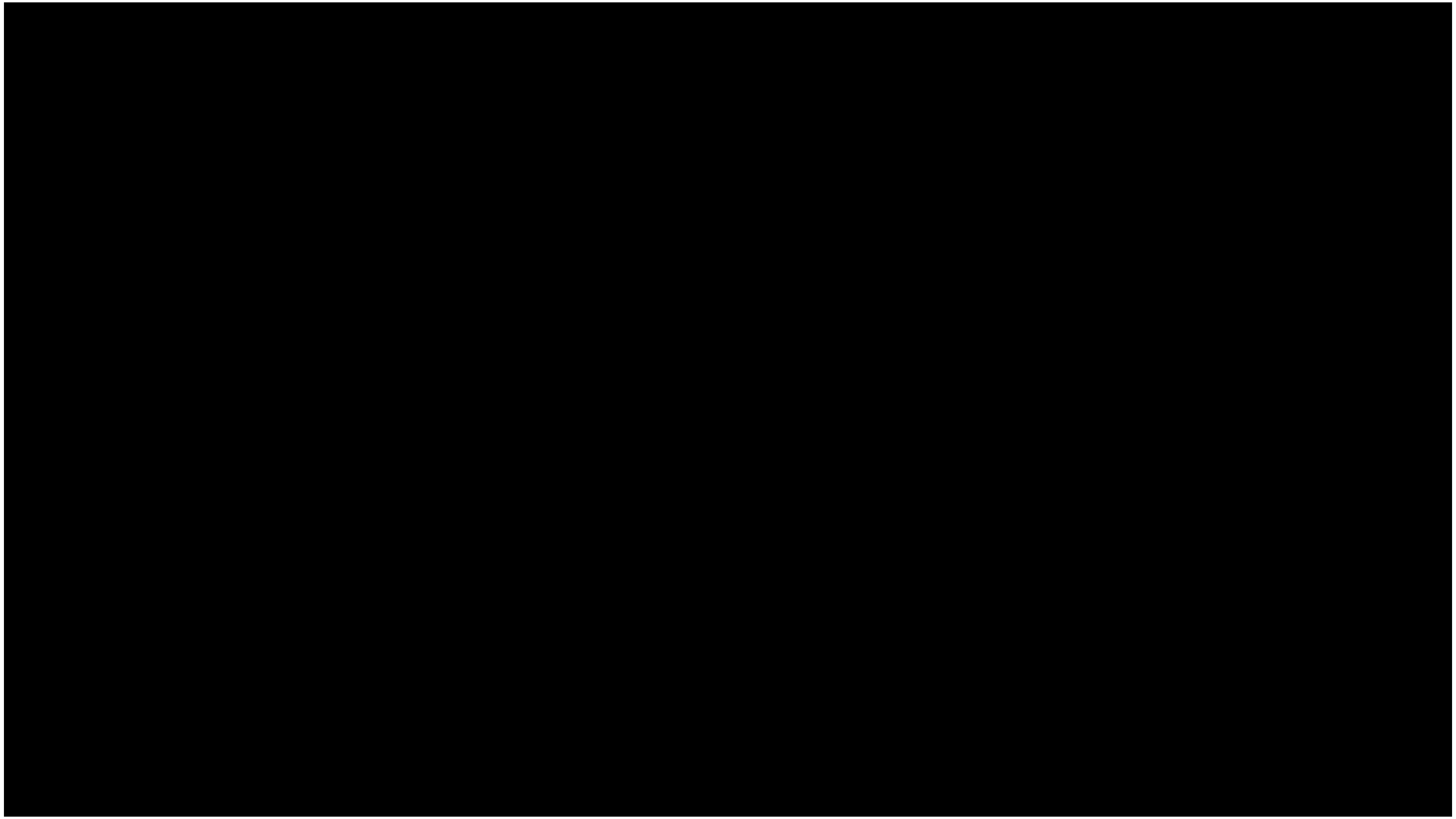
Sorgente delle immagini: https://medium.com/udacity-pytorch-challengers/style-transfer-using-deep-nural-network-and-pytorch-3fae1c2dd73e

AI video generation evolution over years

AI video generation evolution over years

AI video generation evolution over years

Deep Fakes

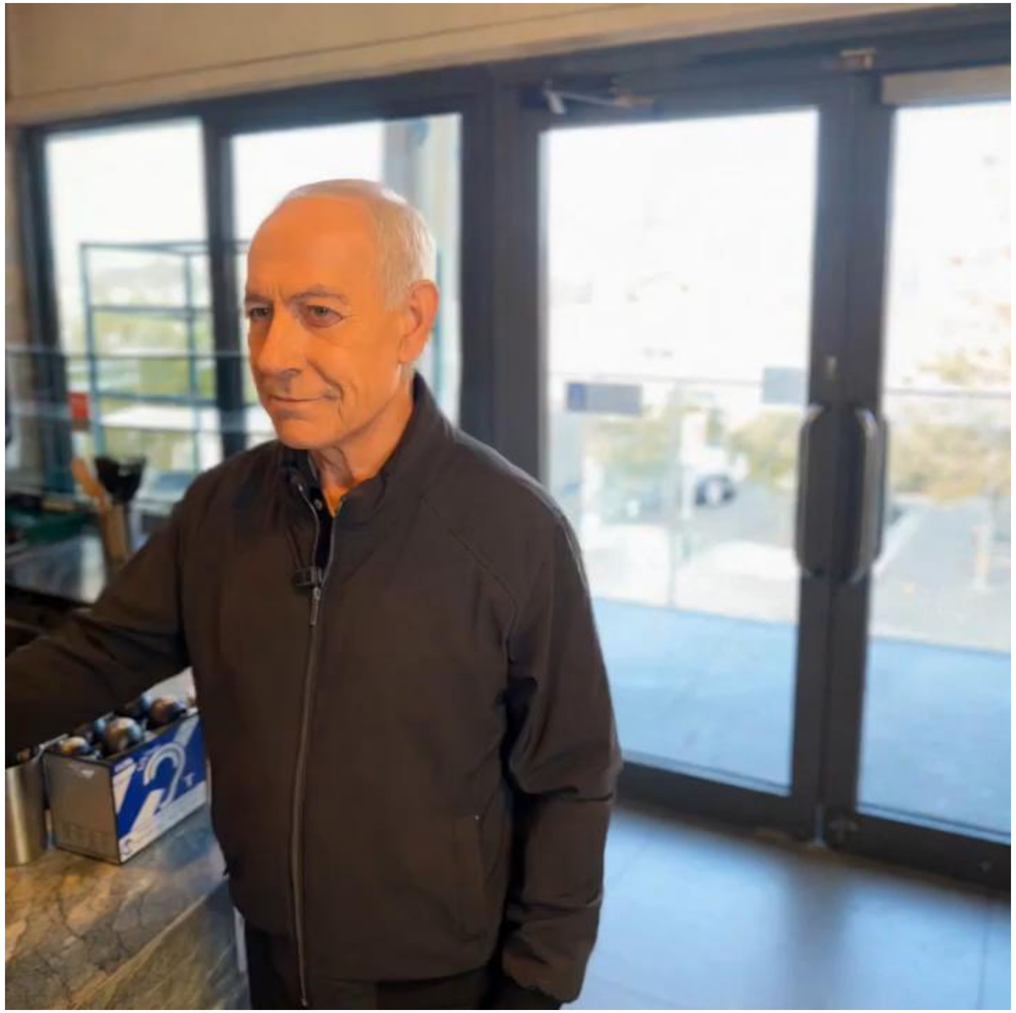


Deep Fakes

Deep Fakes

Deep Fakes

Deep Fakes



Deep Fakes

Deep Fake Detection

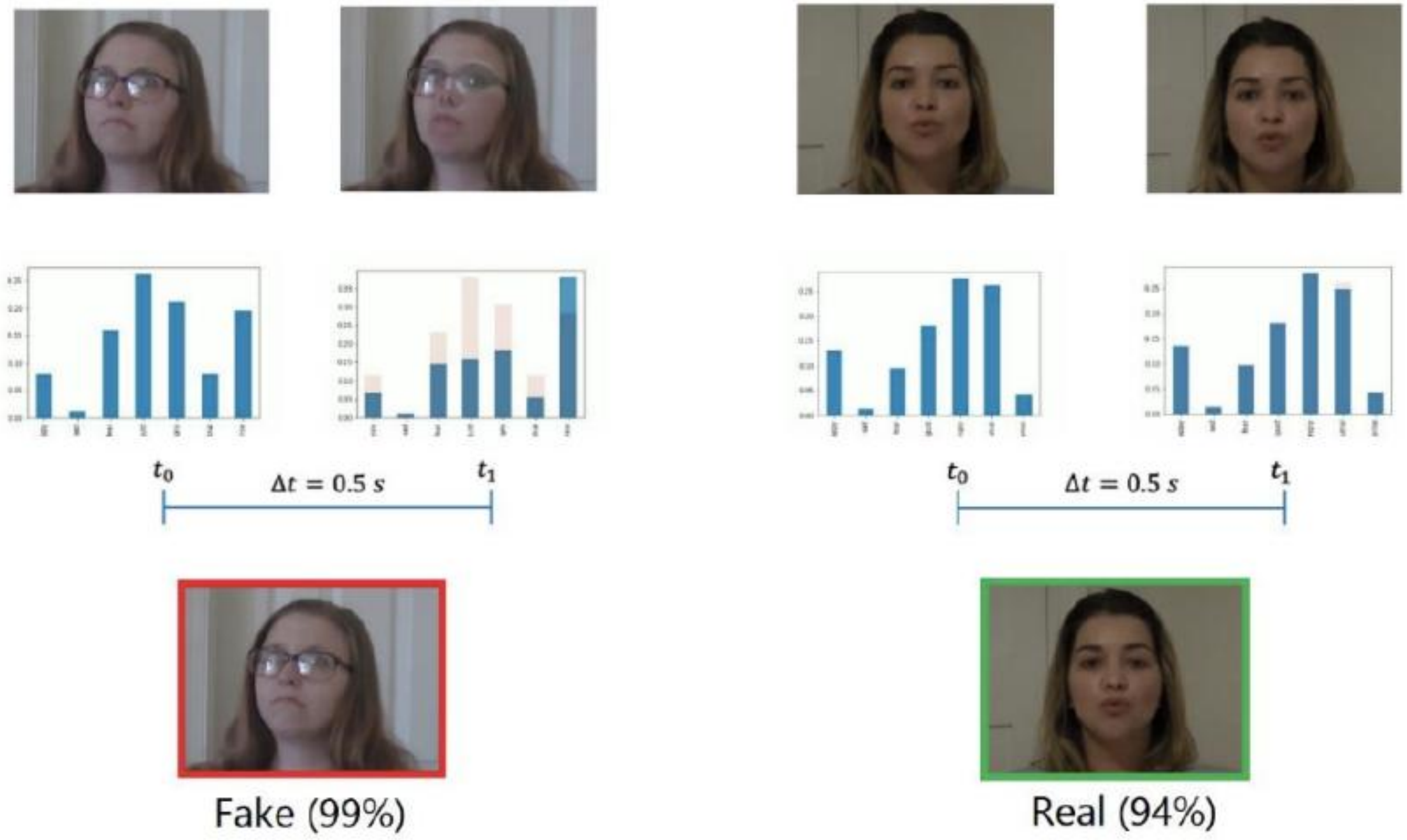
- Deepfakes refer to synthetic media, including images and videos, that are generated using Artificial Intelligence (AI) techniques to alter the appearance or speech of real individuals

- As Deep Learning (DL) approaches have advanced, deepfakes have become increasingly realistic, raising concerns about their potential to spread misinformation and manipulate public opinion
  - ✓ the ability to accurately detect deepfakes has become a critical issue

- One of the main challenges in this field is to identify effective and robust features that can distinguish between real and fake videos.

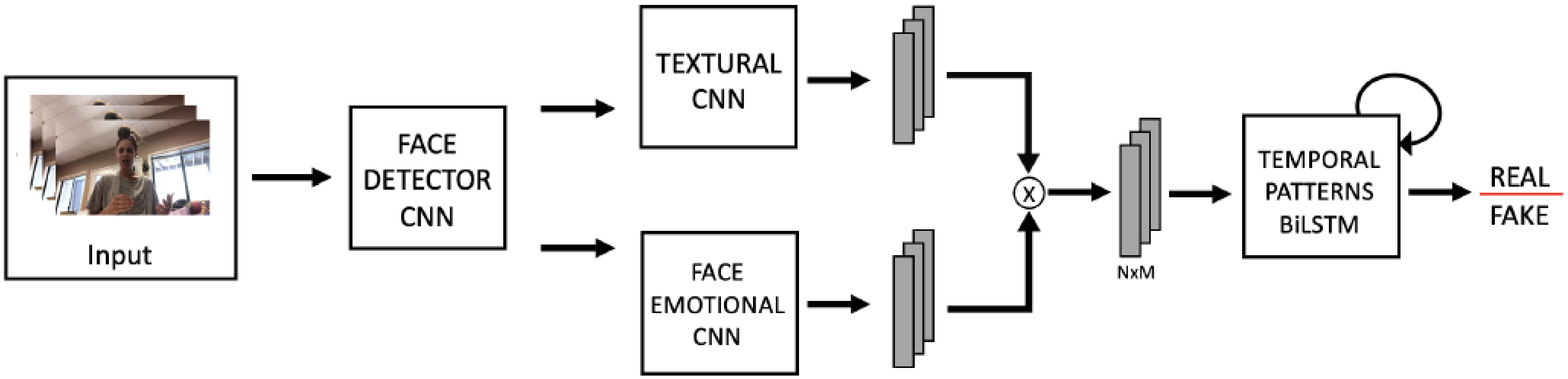
Deep Fake Detection

- To date, not a single approach has been proved effective in all the scenarios
- Emotions have been emerging as a valuable feature for deepfake detection due to the difficulty of synthesizing realistic emotional expressions, which remains a major limitation of deepfake creation algorithms



FEAD-D

- Facial Expression Analysis in Deepfake Detection (FEAD-D, Iscra-C project) aims at exploiting the unnatural variation in the facial expressions introduced by the artefacts generated during the video creation
- The system has been trained and tested on data coming from the Deepfake Detection Challenge (DFDC)
- It exploits Convolutional Neural Networks (CNNs) as features extractors and a bidirectional Long Short-Term Memory (BiLSTM) network to analyse the temporal patterns



Features Extraction

Emotional

- A CNN specifically trained for emotion recognition is used as features extractor
- The network is trained considering data coming from the Facial Expression Recognition (Fer2013) challenge
  - ✓ Seven emotional categories (anger, disgust, fear, happiness, sadness, surprise, and neutral)

Textural

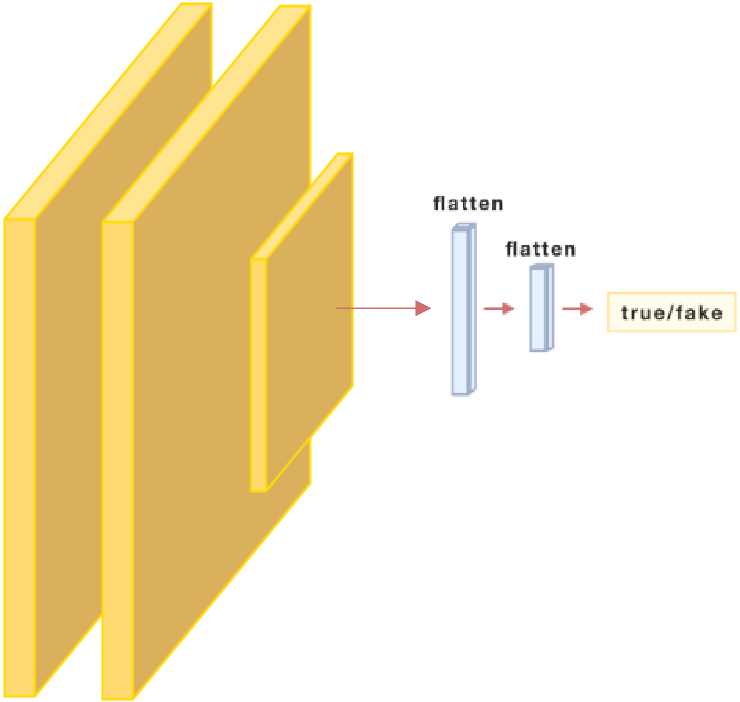
- CNN pre-trained on ImageNet dataset as a textural features extractor
- The aim is to use the textural characteristics for recognizing the artifacts related to the contextual information

- A feature vector is obtained by concatenating the representations produced by the associated CNNs

Temporal Analysis

- The features extracted in the previous stages are analyzed together in a cross-frame fashion to spot incoherent and unnatural patterns in the emotional evolution of the target subject

3 BiLSTM layers

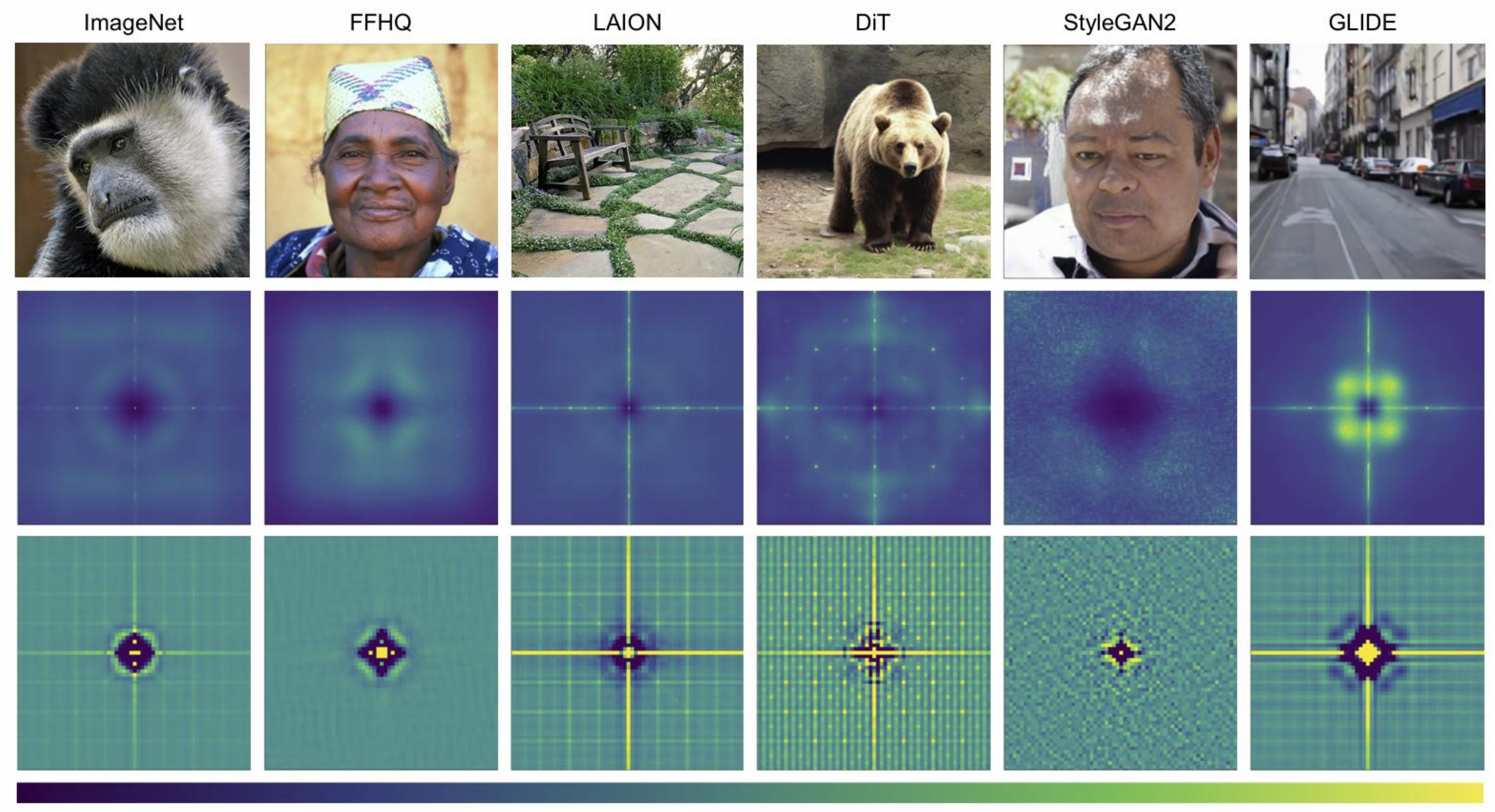


- The resulting system can process a video in two minutes

- It is worth noting that although emotional analysis is a promising approach, it presents **challenges** related to variations across individuals, cultures, and contexts, and the possibility of creating algorithms specifically designed to mimic emotional expressions

Conclusions

Synthetic Image Detection



https://www.grip.unina.it/multimedia-forensics/synthetic-image-detection

Deep Learning ethical threats

- It is clear that the problem is not in ML or CNN, but in humans
- However, ML is been using in several everyday tool (smartphone, ads, loans, etc.)
- How should we manage this?


Joe Redmon @pjreddie · 12h
I stopped doing CV research because I saw the impact my work was having. I loved the work but the military applications and privacy concerns eventually became impossible to ignore.
twitter.com/RogerGrosse/st...
Show this thread

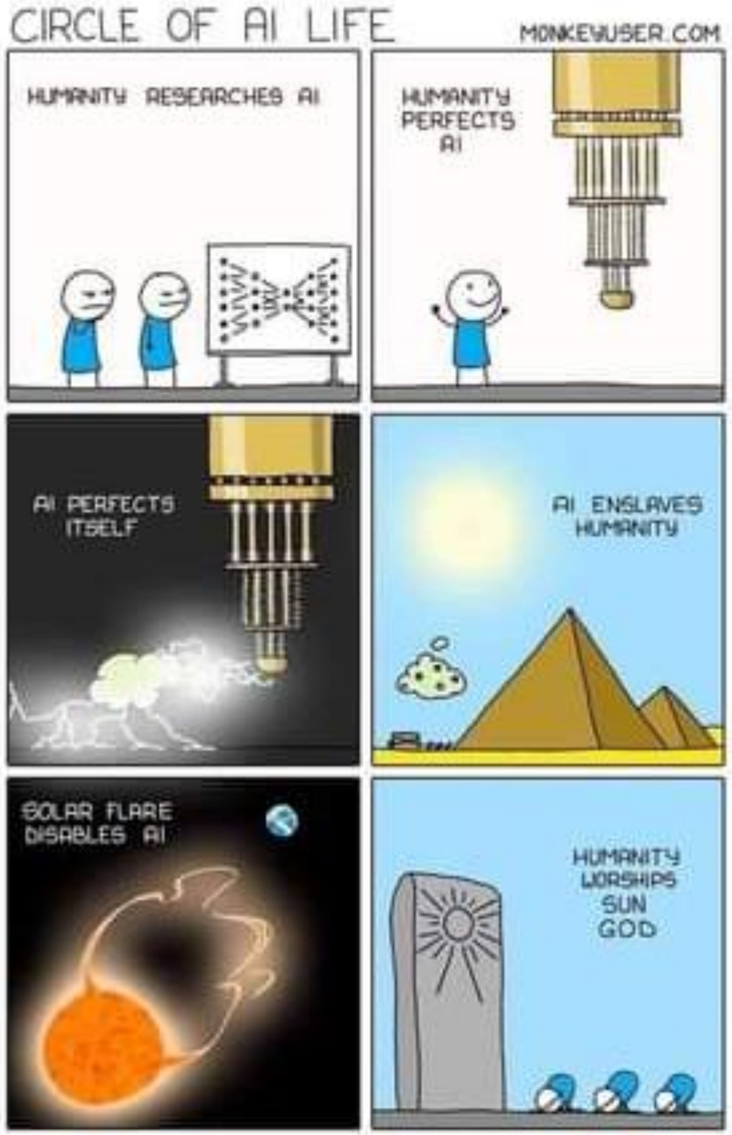

EUROPEAN COMMISSION

Brussels, 19.2.2020
COM(2020) 65 final

WHITE PAPER
On Artificial Intelligence - A European approach to excellence and trust


The Telegraph

Government AI rules to require diverse teams to prevent racist and sexist algorithms



Thanks for your attention!

Questions ?

stefano.marrone@unina.it